

Factivity, presupposition projection, and the role of discrete knowledge in gradient inference judgments*

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Abstract We investigate whether the factive presuppositions associated with some clause-embedding predicates are fundamentally discrete in nature—as classically assumed—or fundamentally gradient—as recently proposed (the Gradient Projection Principle of Tonhauser, Beaver, and Degen 2018). To carry out this investigation, we develop statistical models of presupposition projection that implement these two hypotheses, fit these models to existing inference judgment data aimed at measuring projective inferences (Degen and Tonhauser 2021), and compare the models’ fits to the data using standard statistical model comparison metrics. We find that models implementing the hypothesis that presupposition projection is fundamentally discrete fit the data better than models implementing the hypothesis that it is fundamentally gradient. To evaluate the robustness of this finding, we collect three additional datasets: a replication of the original dataset, as well as two datasets that modify the methodology of the original. Across the three datasets, we again find that models implementing the hypothesis that presuppositions are discrete fit the data better than models implementing the hypothesis that they are gradient. We argue that these results favor an account according to which factive presuppositions are fundamentally discrete in nature, and we discuss how this discreteness might be understood within classical semantic accounts of factive predicates, as well as accounts of presupposition projection that tie it to the question under discussion.

1 Introduction

Semantic theories aim to characterize the inferences that natural language expressions support and to account for at least those inferences which are necessary given the meanings of the expressions. Whether or not a particular inference is necessary is commonly assessed via native speaker judgments. Judgment data, however, tends to be influenced by a number of non-semantic factors. These factors run the gamut: from high-level factors, such as speakers’ prior beliefs about the likelihood that an inference is true, or ambiguities about the relevant expressions, to low-level

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factors, such as the strategies speakers use to map their judgments onto a data collection instrument (e.g., a slider representing likelihood or certainty), or their skill in producing an accurate target response using such an instrument.¹

Testing a semantic theory against inference judgment data therefore requires auxiliary assumptions about the link between (some representation of) these factors and the theoretical constructs of interest. Such linking assumptions are often left implicit in classical methodologies employing informal experiments.² In recent years, however, the need to formulate explicit linking assumptions has become pressing in light of theoretical developments within semantics that have been motivated by finer-grained aspects of the distribution of inference judgments than can be observed informally.

One domain where large collections of inference judgments have become particularly important is presupposition projection and, in particular, factivity. A predicate is said to be *factive* if it is implicated in triggering *veridicality inferences* (i.e., inferences that its embedded clause is true) regardless of whether or not certain entailment canceling operators take scope over the predicate (Kiparsky and Kiparsky 1970). For example, *love* is often taken to be factive, since sentences such as those in (1) give rise to the inference in (2).

- (1) a. Jo loves that Mo left.
 b. Jo doesn't love that Mo left.
 c. Does Jo love that Mo left?
 d. Jo might love that Mo left.
 e. If Jo loves that Mo left, she'll also love that Bo left.

- (2) Mo left.

In particular, the inference in (2) is said to *project* throughout (1). Diagnosing factivity based on inference judgments is known to be challenging, since inferences are greatly influenced by a predicate's context of use (Karttunen 1971 *et seq.*). To better understand the factors driving factive inferences, many researchers have begun to collect judgments from native speakers in formal experiments, often in large quantities, in order to evaluate hypotheses about the semantic properties of potentially factive predicates, as well as about how these semantic properties are reflected in the distributional properties of judgment data (Tonhauser 2016; Djärv and Bacovcin 2017; Djärv, Zehr, and Schwarz 2018; White and Rawlins 2018b; White, Rudinger, et al. 2018; White 2021; Degen and Tonhauser 2021; Degen and Tonhauser 2022; Jeong 2021; Kane, Gantt, and White 2022).

Of particular concern in the experimental literature on factivity has been the observation that, in tasks aimed at measuring a predicate's factivity, aggregate measures of inference judgments show much more gradience than one might initially expect under a classical view of factivity as

¹We refer to as “high-level” those factors which we take to be underlie linguistic inference (e.g., the semantics of expressions), and as “low-level”, those we take to constrain the specific task (e.g., cognitively or physically) in which the linguistic inferences of interest are hypothesized to be implicated.

²See Jasbi, Waldon, and Degen 2019 and Waldon and Degen 2020 for recent approaches to modeling linking hypotheses in formal inference and truth value judgment experiments.

a discrete and invariable lexical property (White and Rawlins 2018b). Some authors have gone so far as to claim that the observed gradience indeed casts doubt on this view (Degen and Tonhauser 2022). Doubt about the existence of such a lexical property is furthermore consistent with the view that presupposition projection is fundamentally gradient in general (Tonhauser, Beaver, and Degen 2018). This *Fundamental Gradience Hypothesis* contrasts with a *Fundamental Discreteness Hypothesis*, under which factivity comprises a discrete property or collection of properties of an expression on a particular occasion of use. We discuss these hypotheses in more detail in Section 2.

Our central aim in this paper is to quantitatively evaluate the two hypotheses mentioned above in terms of a framework that allows us to explicitly formulate the link between their respective construals of factivity and the way humans produce inference judgments. The core theoretical contribution we make in employing this framework, which is developed in Grove and White 2024, is to transparently relate the sorts of compositional analyses of the meanings of factive predicates common in the formal semantics literature to probabilistic models characterizing distributions over inference judgments.

We outline the main ingredients of this framework in Section 3 before using it to carry out a comparison between the Fundamental Gradience Hypothesis and the Fundamental Discreteness Hypothesis in Sections 4 and 5 (see Grove and White 2024 for an introduction to the framework in its full generality). As we show, the framework of Grove and White allows us to precisely target where the two hypotheses make different predictions about the distribution of inference judgments across participants, making an apples-to-apples comparison feasible. Moreover, it allows us to do so using standard statistical model comparison metrics that balance out a model's fit to inference judgment data against the model's complexity.

Using such metrics, we find that models that implement the Fundamental Discreteness Hypothesis unambiguously outperform models that implement the Fundamental Gradience Hypothesis, across both an existing dataset aimed at measuring projective inferences (Degen and Tonhauser 2021) and three new datasets: a replication of the existing dataset as well as two new datasets that modify Degen and Tonhauser's paradigm to evaluate model robustness. In Section 6, we argue that these results favor an account according to which factive presuppositions are fundamentally discrete in nature, and we discuss how this discreteness might be understood within classical semantic accounts of factive predicates, as well as accounts that attempt to reduce presupposition projection to entirely pragmatic processes (Simons 2007; Simons, Tonhauser, et al. 2010; Simons, Beaver, et al. 2017).

2 Gradient inference patterns among factive predicates

Inference judgment datasets—such as MegaVeridicality (White and Rawlins 2018b; White, Rudinger, et al. 2018), VerbVeridicality (Ross and Pavlick 2019), and CommitmentBank (De Marneffe, Simons, and Tonhauser 2019)—has enabled fine-grained analyses of inference judgment patterns across the entire clause-embedding lexicon. Remarkably, the aggregate judgments of multiple speakers seen in such datasets reveal substantial gradience, a fact which has garnered sustained focus in recent work.

In the domain of factivity, White and Rawlins (2018b) observe gradience in the aggregate measures of various predicates' degrees of factivity, using data from the MegaVeridicality dataset (summarized in Figure 1; see also Xue and Onea 2011; Smith and Hall 2011; Djärv and Bacovcin

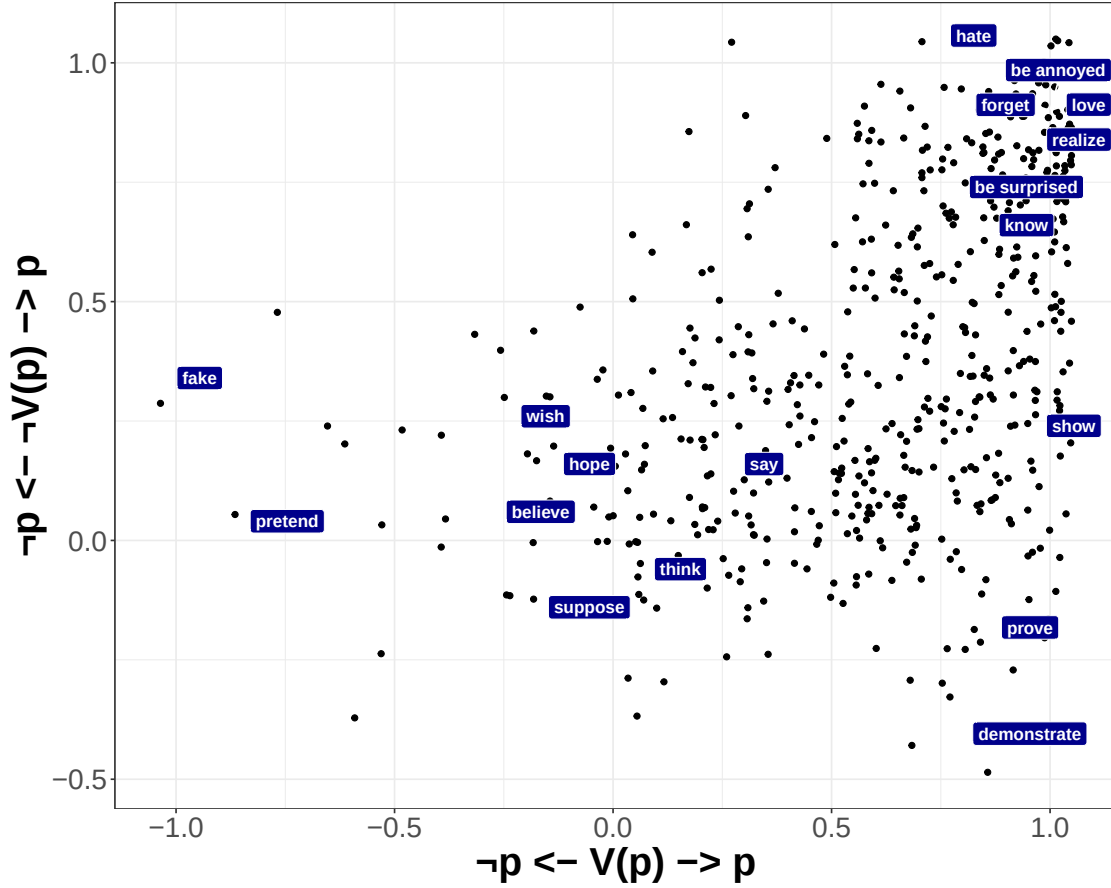


Figure 1: An aggregate measure of factivity, derived from the MegaVeridicality dataset of White and Rawlins 2018b. Points are jittered (0.05) to avoid overplotting. The x -axis corresponds to the mean scores, across participants, of responses to prompts of the form *Someone {didn't, wasn't} __ that a particular thing happened. Did that thing happen?* with the same possible responses. Each point is a predicate, with a selection of points labeled. Note that jittering sends some of the points above or below 1 and -1, respectively.

2017 for related observations). Crucially, they note a lack of clear dividing lines among classes of predicates and suggest that speakers' gradient aggregate veridicality ratings are likely influenced by the fine-grained semantics of particular verbs (*ibid.*, p. 228; cf. Roberts and Simons 2024).³

In later work, Degen and Tonhauser (2022) investigate the nature of inferential gradience in six experiments and argue that its persistence rebuts the hypothesis that there is a coherent class of factive predicates. Our own modeling work uses data collected under the same experimental paradigm as that which Degen and Tonhauser employ, so we describe their data and arguments in detail in Section 2.1. In Section 2.2, we turn to the broad question of which factors may be responsible for the gradience observed in inference judgment tasks. We outline an array of explanatory choices that one can make in attempting to account for this gradience, and we discuss some of the evidence that would be pertinent to these choices. We then present two alternative hypotheses about the nature of gradience among inference judgments that constitute the main focus of this paper—the Fundamental Discreteness Hypothesis and the Fundamental Gradience Hypothesis—and we relate them to the kinds of explanatory choices we have introduced. In Section 2.4, we discuss the specific paradigm we use to evaluate these two hypotheses.

2.1 Measuring veridicality and factivity

In all of their experiments, Degen and Tonhauser (2022) focus on the set of twenty clause-embedding predicates listed in (3), which they group into hypothetical classes based on the prior literature on factivity (Kiparsky and Kiparsky 1970; Karttunen 1971; Hooper and Thompson 1973; Givón 1973; Hooper 1975; Abusch 2002; Abusch 2010; Abrusán 2011; Abrusán 2016; Anand and Hacquard 2014, i.a.).

(3) Twenty clause-embedding predicates (Degen and Tonhauser 2022, p. 559, ex. 13)

- a. canonically factive: *be annoyed, discover, know, reveal, see*
- b. non-factive
 - (i) non-veridical non-factive: *pretend, say, suggest, think*
 - (ii) veridical non-factive: *be right, demonstrate*
- c. optionally factive: *acknowledge, admit, announce, confess, confirm, establish, hear, inform, prove*

In their discussion of the experimental data which they go on to collect, they suggest that insofar as the standard classification of predicates is correct, "... we expect to see a categorical difference in projection between canonically factive predicates on the one hand, and optionally factive and nonfactive predicates on the other" (p. 569). To measure projection, Degen and Tonhauser provide participants with a scenario in which someone asks a polar question whose main verb is one of the clause-embedding predicates of interest, e.g., (4).

³This gradience is not White and Rawlins's main focus; they are primarily interested in the relationship between the veridicality inferences associated with particular predicates and those predicates' syntactic distributions (as opposed to their semantic classification), and that relationship can be assessed without resolving what drives the gradience they observe in the aggregated inference judgments.

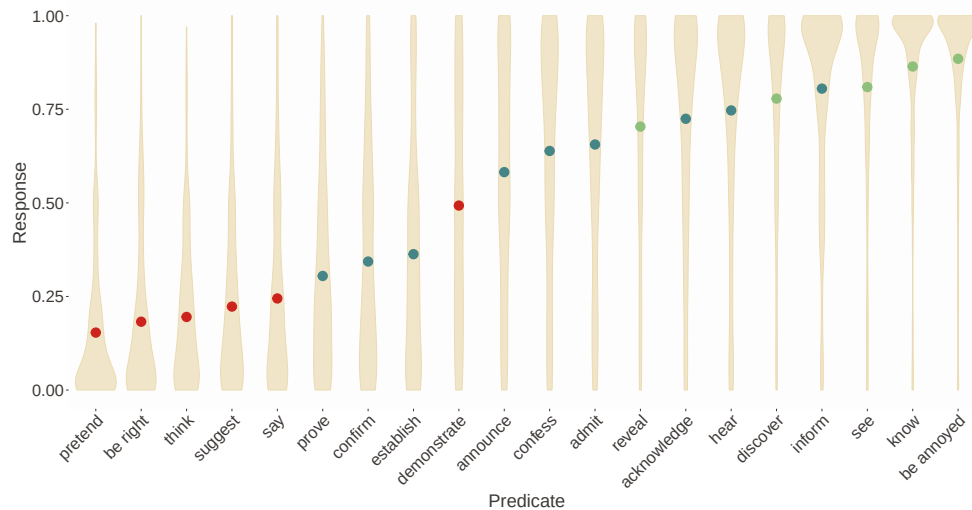


Figure 2: Verb means from Degen and Tonhauser’s (2022) experiment 1a. “Non-factive” verbs are in red, “optionally factive” verbs are in teal, and “canonically factive” verbs are in green. Violin plots indicate the probability density of responses. (See Degen and Tonhauser 2022, Figure 2, p. 562.)

(4) **Helen asks:** Did Amanda discover that Danny ate the last cupcake?

They then ask participants to provide a rating on a continuous slider scale from *no* to *yes* in answer to a prompt of the form in (5). This prompt is meant to assess the extent to which participants believe that the content of the embedded clause projects, i.e., by using speaker certainty as a proxy for projection.

(5) Is Helen certain that Danny ate the last cupcake?

In another experiment, they give participants a variant of this task in which their answer is provided as a binary forced choice between *no* and *yes*, rather than a slider scale response.

Degen and Tonhauser also claim that categories of predicates ought to emerge when one analyzes judgments of veridicality: “we expect the [contents of the complements of] canonically factive and veridical nonfactive predicates to be entailed” (p. 569). They assess veridicality inferences using two methods. First, they provide participants with a description of a scenario in which a sentence containing one of the predicates of interest is assumed to be true, as in (6).

(6) **What is true:** Edward proved that Grace visited her sister.

They follow this scenario description with a prompt of the form in (7).

(7) Does it follow that Grace visited her sister?

Depending on the experiment, participants either answer on a sliding scale whose endpoints are labeled *no* and *yes*, or they are asked to make a binary forced choice between *no* and *yes*.

Second, Degen and Tonhauser give participants a description of a scenario in which someone makes an utterance which should be contradictory if the relevant complement clause is entailed, as in (8).

(8) **Margaret:** “Edward heard that Mary is pregnant, but she isn’t.”

Participants are then prompted to answer a question of the form in (9), either on a sliding scale or by making a binary forced choice, depending on the experiment.

(9) “Is Margaret’s utterance contradictory?”

Consistent with White and Rawlins’s original observations, Degen and Tonhauser find that the patterns of inference ratings across predicates in their experiments are gradient in nature for both projection and veridicality inferences. Indeed, the degree to which predicates display projective inferences appears to evolve continuously from the least projective predicate (*pretend*) to the most projective predicate (*be annoyed*) when predicates are compared in terms of their mean ratings (see Figure 2). Such gradience is manifest in both of the experiments assessing projection: the one which collects sliding scale judgments and the one which collects binary judgments. A similar pattern emerges in the experiments assessing veridicality inferences. Crucially, no predicate patterns consistently with the veridical control items across all four experiments assessing veridicality.

2.2 Gradience in inference datasets

Degen and Tonhauser’s results are consistent not only with White and Rawlins’s original observations, but also with findings from adjacent domains. An and White (2020) observe similar gradience in neg-raising inferences, as captured in their MegaNegRaising dataset. Meanwhile, Kane, Gantt, and White (2022) note a similar pattern among the belief and desire inferences they obtain in their MegaIntensionality dataset.

Kane, Gantt, and White note that in the face of such gradience, it is reasonable to entertain two kinds of hypotheses. On the one hand, it is possible that “apparent gradience indicates that no formally represented lexical property controls whether a particular inference is triggered” (p. 572). On the other hand, “apparent gradience [may be] partly or wholly a product of the methods often used to collect inference judgments, and [it may be] that there are discrete, formally represented lexical properties that are active in triggering... inferences” (p. 572). To pursue this question, they ask whether clear *patterns* emerge across the inference judgment datasets discussed above—MegaVeridicality, MegaNegRaising, and MegaIntensionality—by clustering predicates into classes sensitive to the responses from those datasets so as to optimize their ability to predict predicates’ syntactic distributions, as measured in the MegaAcceptability dataset (White and Rawlins 2016).⁴ They uncover fifteen classes that correspond extremely closely to those that one would expect from prior work on clause-embedding predicates. As shown in Figure 3, these classes include a variety of factive subclasses that differ principally in the patterns of belief and

⁴The idea behind optimizing the predictability of predicates’ syntactic distributions is that insofar as the classes to which a predicate belongs are predictive of the predicate’s syntactic distribution, there is preliminary evidence that such classes are associated with distinct distributionally active lexical representations.

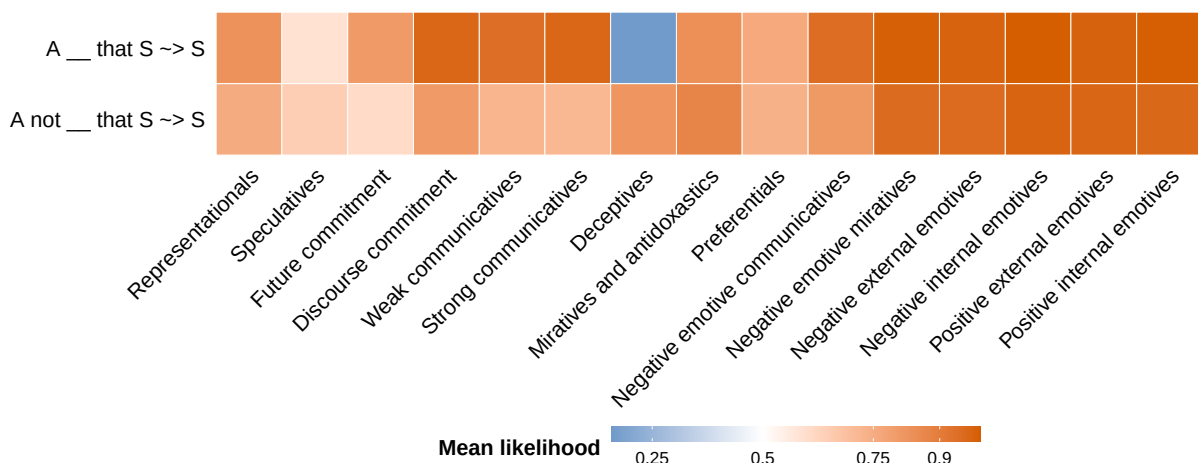


Figure 3: Veridicality inferences associated with each class found in Kane, Gantt, and White 2022. Kane, Gantt, and White provide labels for each class based on the predicates that occur in that class as well as the belief and desire inferences associated with that class. The top row and bottom row correspond to the x - and y -axes of Figure 1, respectively. The classes associated with dark orange cells in both rows are taken to be factive subclasses.

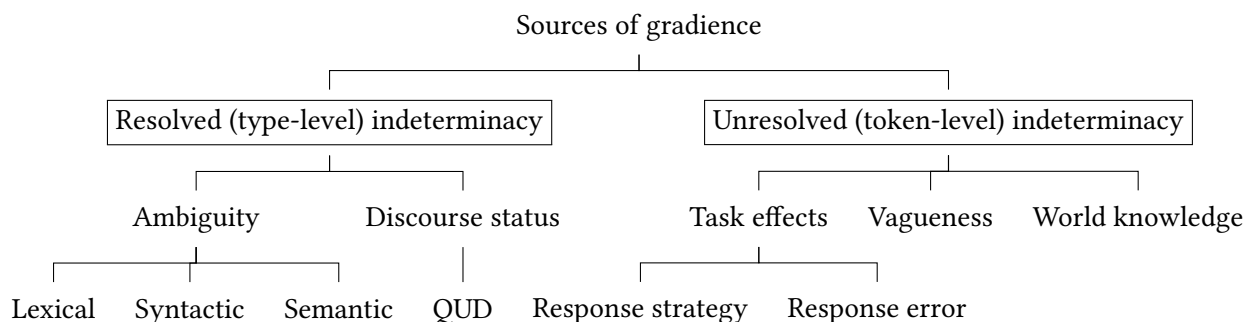


Figure 4: A partial taxonomy of the sources of gradience in inference judgment tasks.

desire inferences they are associated with. As one might expect (Karttunen 1971 *et seq.*), the true factive subclasses tend to be emotive and include, for example, *love* and *hate*.

Kane, Gantt, and White’s findings therefore establish that there *is* a coherent class of factive predicates—the *emotives*—which are, in turn, subclassed by the belief and desire inferences they give rise to. But they also find that there are a variety of classes associated with weaker veridicality inferences than one might expect from a truly factive class. These classes include non-emotive predicates, like *know* and *realize*. Hence, while it is not correct to say that there is *no* class of factive predicates, one must still explain the apparent gradience associated with certain classes, such as the non-emotive ones, as Degen and Tonhauser point out (their Section 4.1, Objection 3). Class-level gradience of this kind is unlikely to be—as Kane, Gantt, and White put it—“partly or wholly a product of the methods often used to collect inference judgments”, since their analysis expressly accounts for the relevant task effects.

2.3 Sources of gradience

How, then, should one account for gradience, whether it is due to knowledge about particular predicates or the classes those predicates fall into? To approach this question, we regiment the possible sources of gradience into two broad classes: *resolved* (or *type-level*) indeterminacy and *unresolved* (or *token-level*) indeterminacy. A partial taxonomy of these sources is provided in Figure 4.⁵

2.3.1 Resolved indeterminacy

Resolved indeterminacy drives uncertainty one might have about the *type* of speech act one is drawing an inference from. Consider an uncontroversial example of such indeterminacy: in an utterance of the sentence *my uncle is running the race*, the verb *run* is ambiguous; the speaker's uncle might be a participant in the race, or he might be in charge of its logistics. If one is presented with an utterance of this sentence and is then asked, "On a scale of 1 to 10, how likely is it that my uncle has good managerial skills?", one might be tempted to pick a lower number—say, around 2—if *run* is interpreted in its locomotive sense, while one might be tempted to pick a higher number—say, around 8—if *run* is interpreted in its managerial sense. Indeed, if a hundred people are presented with an utterance of this sentence and are then asked the same question, perhaps about half of them will interpret *run* in the locomotive sense, while the other half interpret it in the managerial sense. Thus, while the distribution of judgments will look bimodal—with one mode around 2 and the other mode around 8—the *average* judgment will be around 5. Gradience that stems from a semantic ambiguity—i.e., a type of resolved indeterminacy—is thus observable at the level of an entire experiment (i.e., an inference judgment dataset), rather than at the level of an individual trial (a single row of such a dataset).

Other presumably uncontroversial examples of resolved indeterminacy involve semantic ambiguities relating to, e.g., quantifier scope and binding, or (possibly more controversially) discourse-level properties, such as the question under discussion (QUD; Ginzburg 1996; Roberts 2012). Each may lead to distinct, all-or-nothing inferences produced on different utterance occasions.

2.3.2 Unresolved indeterminacy

Unresolved indeterminacy drives uncertainty that persists about the status of an inference even after having fixed the meanings of the expression being used to perform a particular speech act. One such kind of uncertainty is associated with *vagueness*; for example, from an utterance of *my uncle is tall*, there can be uncertainty about whether or not my uncle is at least six feet tall. Another comes from *world knowledge*; given an utterance of *my uncle is running the race* (and having fixed *run* to its locomotive sense), there can still be uncertainty about how fast he runs, given that he is a runner.

Crucially, such uncertainty would be reflected on *individual trials* of an inference judgment experiment. Moreover, resolved indeterminacy and unresolved indeterminacy can coexist: in the

⁵Indeed, this taxonomy is only partial (and admittedly, conjectural). For instance, as Chris Kennedy (p.c.) points out, expressions displaying *open texture*—the pattern of uncertainty about the extensions of, e.g., non-gradable predicates such as *sandwich* (is a hot-dog a sandwich?)—seem likely to give rise to gradience (Mackinnon, Waismann, and Kneale 1945), but it is potentially contentious whether open texture should be thought of as type- or token-level indeterminacy.

hypothetical inference judgment task described above, there is uncertainty about the intended sense of the verb *run*, and there is uncertainty about the inference that the referent of *my uncle* has good managerial skills, which itself stems from world knowledge.

Finally, unresolved indeterminacy in an experimental setting might also stem from task effects, such as one's physical response error or one's response strategy; for example, some experimental participants might hedge their responses so that they are never quite at the extremes of the relevant measurement scale. This indeterminacy is unresolved in the sense that it remains even after fixing the interpretation of an expression.

2.3.3 The application to factivity

In the realm of factive inferences, one could adopt the view that gradience among individual predicates and classes of predicates is unresolved in nature. This view would align with Tonhauser, Beaver, and Degen's *Gradient Projection Principle* (ex. 7, p. 499):

- (10) **Gradient Projection Principle:** If content *C* is expressed by a constituent embedded under an entailment-canceling operator, then *C* projects to the extent that it is not at-issue.

Alternatively, one might adopt the view that such indeterminacy is actually resolved. As Degen and Tonhauser put it, the latter view says that “the observed gradience in projection [is] compatible with a binary factivity category in combination with two assumptions: first, that predicates may be ambiguous between a factive lexical entry... and a nonfactive lexical entry... and, second, that interpreters may be uncertain about which lexical entry a speaker intended in their utterance” (p. 583). Our task in this paper is to formalize these two types of explanation so that we may quantitatively compare them.

We thus consider two hypotheses about the sources of gradience in projective inference judgments among clause-embedding predicates:

- (11) a. *The Fundamental Discreteness Hypothesis.* Factivity is a discrete property of at least some token occurrences of expressions containing at least some clause-embedding predicates. A given use of an expression containing a clause-embedding predicate either triggers a projective inference, or it does not trigger a projective inference on a particular token occurrence.
- b. *The Fundamental Gradience Hypothesis.* There is no property distinguishing factive from non-factive occurrences of clause-embedding predicates. Instead, the gradient distinctions observed among predicates and their classes reflect different gradient contributions different predicates make to inferences about the truth of their complement clauses.

According to the Fundamental Discreteness Hypothesis, the gradience associated with the predicates in (3) is due to resolved indeterminacy. Different occasions on which a predicate is used may vary with respect to whether or not the expression containing the predicate triggers a projective inference, but such variation concerns which of the alternative interpretations of the expression should be selected. Alternative interpretations may be available for a variety of reasons: among

other possibilities, (i) a predicate may have multiple senses—at least one that is implicated in triggering projection and at least one that is not; (ii) a predicate may occur in multiple structures—at least one that is implicated in triggering projection and at least one that is not;⁶ or (iii) a predicate may be usable in contexts where the common ground—or some other construct, such as the question under discussion—entails the content of its embedded clause, as well as contexts where it does not (see Simons, Beaver, et al. 2017; Roberts and Simons 2024).⁷ The gradience associated with particular classes of predicates that Kane, Gantt, and White observe might then indicate (respectively) (i) that there is a sort of regular polysemy among predicates within a class; (ii) that predicates within a class bias the resolution of syntactic ambiguity in similar ways; or (iii) that predicates within a class tend to show up in similarly structured contexts. It is noteworthy that possibility (iii) may be seen to be instantiated by the *Projection Principle* of Beaver et al. (2017, ex. 33, p. 280), which one might interpret as a *discrete* variant of the Gradient Projection Principle:

- (12) **Projection Principle:** If content C is expressed by a constituent embedded under an entailment-canceling operator, then C projects if and only if C is not at-issue.

On this view, certain (classes of) predicates may simply tend to show up in contexts in which the contents of their complements are (or aren't) at-issue. With this in mind, we stress that the Fundamental Discreteness Hypothesis is not a hypothesis about whether the indeterminacy that drives the observed gradience is relevant to any particular semantic or pragmatic construct; it is about the nature of the resolution of that indeterminacy.

According to the Fundamental Gradience Hypothesis, observed gradience is a product of unresolved indeterminacy: it is more like vagueness than it is like ambiguity. This hypothesis is consistent with Tonhauser, Beaver, and Degen's (2018) Gradient Projection Principle. It is important to note that Tonhauser, Beaver, and Degen (2018) themselves consider two interpretations of this principle. On the one hand, they suggest that "a listener's (or reader's) judgment that a content is projective to a certain extent means that the listener takes the speaker (or writer) to be committed to the content to that extent" (p. 498); and on the other, they consider that "a listener's judgment that a content is projective to a certain extent reflects the probability with which they believe the speaker to be committed to the content" (pp. 498–499).

The first of these interpretations is unambiguously consistent with our Fundamental Gradience Hypothesis. The second interpretation has a more ambiguous status and might, in principle, be seen as consistent with either of our hypotheses. As Tonhauser, Beaver, and Degen say, "[o]n this interpretation, speaker commitment may be a binary, categorical property and projection variability arises from the listener's uncertainty about the whether the speaker is committed" (p. 499). That is, Tonhauser, Beaver, and Degen consider the possibility that a speaker's commitment might be discrete, while the uncertainty that informs a listener's judgment may be gradient; e.g., if it is encoded as a probability. This possibility might be viewed as consistent with the Fundamental Discreteness Hypothesis insofar as it allows that listeners end up making a judgment

⁶The second option is plausible in light of cross-linguistic evidence that functional items surrounding a predicate can modulate veridicality inferences; e.g., nominal morphology attached to verbs or their clausal complements (see Varlokosta 1994; Giannakidou 1998; Giannakidou 1999; Giannakidou 2009; Roussou 2010; Farudi 2007; Abrusán 2011; Kastner 2015; Ozyildiz 2017; see White 2019 for discussion).

⁷See Qing, Goodman, and Lassiter 2016 for an approach within the Rational Speech-Act framework (Frank and Goodman 2012; Goodman and Stuhlmüller 2013) along the lines of possibility (iii).

about whether they take the speaker to be committed to the content of the embedded clause in spite of their uncertainty, rather than retaining that uncertainty as the discourse develops.

For the purposes of this paper, we interpret the Gradient Projection Principle to imply a view of gradience in the patterns of inference across predicates as arising from unresolved indeterminacy. We believe that this interpretation is reasonable in light of the fact that the alternative one, under which the Gradient Projection Principle is consistent with our Fundamental Discreteness Hypothesis, would render it fairly vacuous: under this interpretation, the Gradient Projection Principle reduces to the vanilla Projection Principle.

2.4 The role of world knowledge in gradient inference judgments

Our tests of these hypotheses will rely crucially on a paradigm used by Degen and Tonhauser (2021), who aim to characterize the influence of world knowledge on projective inferences, focusing on the same twenty clause-embedding predicates as in (3). We employ this paradigm because it allows us to explicitly model the influence of non-semantic factors—specifically, speakers’ prior beliefs about the likelihood of an inference being true. Similar to the experiment reported above, in which Degen and Tonhauser (2022) measure projection out of the complement of a predicate placed inside of a polar question, Degen and Tonhauser (2021) measure projective inferences in the presence of a background fact whose content they manipulate (their experiment 2b). To illustrate, the following experimental trial features the predicate *pretend*:

Fact (which Elizabeth knows): Zoe is a math major.
Elizabeth asks: "Did Tim pretend that Zoe calculated the tip?"

Is Elizabeth certain that Zoe calculated the tip?

noyes

Next

The same twenty complement clauses as featured in Degen and Tonhauser 2022 are also featured in this experiment, but now each clause is paired with one of two background facts, constructed so that one fact makes the clause more likely to be true than the other does. Each participant in this experiment sees twenty items (along with six control items). On each experimental trial, a predicate is placed in the context of one of the twenty clauses, along with one of the two background facts constructed for that clause. The results Degen and Tonhauser (2021) obtain in this setting mirror those of Degen and Tonhauser (2022). In particular, the mean projection ratings for the twenty predicates show a similar gradient pattern (Spearman’s $r = 0.98$).

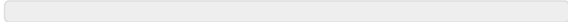
In addition to the assessment of projective inferences given background facts, Degen and Tonhauser (2021) conduct a norming experiment, in which the prior certainties about the truth of the complement clauses featured in their projection experiment are assessed independently, given the same background facts (their experiment 2a). Trials in this experiment ask participants

to judge how likely the relevant clause is to be true, given one of the two background facts constructed for it. For example, the following trial features the same clause as in the example provided previously, but with the alternate low-probability fact:

Fact: Zoe is 5 years old.

How likely is it that Zoe calculated the tip?

impossible
definitely



Degen and Tonhauser find that the by-item means for the forty pairs of complement clauses and background facts, as assessed in their norming experiment, are a good linear predictor of the inference ratings for items featuring the same complement clauses and facts which they obtain in their experiment investigating projective inferences.

Thus, at least one source of the variation among the projective inferences associated with clause-embedding predicates is the *context* in which these predicates are placed; and in particular, the prior certainties that people associate with these contexts. This cannot be the whole story, of course, as Degen and Tonhauser observe: the mean projection ratings for predicates display substantial gradience even after collapsing across the contexts in which they occur (consistent with their 2022 experiment 1a: see Figure 2). So, what explains the remaining variation?

In the next section, we introduce a semantic and pragmatic framework that allows us to precisely state *theories* of gradience and to rigorously relate these theories to probabilistic models of inference judgment data. When constructing such theories, we attempt to remain relatively abstract, and to include only as much semantic detail as we need in order to distinguish the two hypotheses in (11). At the same time, we aim to be clear about how one could elaborate on our account, as implemented, in order to spell out more fine-grained variants of the hypotheses.

3 Probabilistic semantics

To state a theory of gradience precisely, it is useful to have a general method of integrating probabilistic reasoning into a compositional semantics of English. Here we rely on the framework of Probabilistic Dynamic Semantics (PDS; Grove and White 2024), which follows earlier work by Grove and Bernardy (2023) in adopting an interface for performing Bayesian reasoning in typed λ -calculus using *monads*.⁸ Our reason for employing this framework is that it allows one to transparently relate the sorts of compositional analyses of expressions' meanings common in formal semantics to probabilistic models that characterize distributions over inference judgments.

Importantly, the framework allows the two hypotheses laid out above to be precisely specified, while keeping fixed both the formal analysis of the expressions of interest and the way in which probability distributions over inference judgments are mapped onto a particular data collection

⁸See Giorgolo and Asudeh 2014; Asudeh and Giorgolo 2020 and Bernardy et al. 2018 for related monadic approaches. The aforementioned works have somewhat different aims from Grove and Bernardy's, which are reflected in the distinct interfaces they supply.

instrument. These aspects of the framework are important because (as we show in Sections 4 and 5) they allow us to conduct an apples-to-apples comparison of the two hypotheses which both (a) precisely targets where they make different predictions about the distribution of inference judgments across experimental trials, and (b) does so using standard statistical model comparison metrics which balance out a model’s fit to inference judgment data against its complexity.

We begin in Sections 3.1 to 3.5 with relevant background on Grove and White’s framework before turning in Section 3.6 to an example. Our aim is to show how one may finely delineate indeterminacy that is core to the semantic value of an expression—giving rise to unresolved indeterminacy—from indeterminacy about which interpretation should be associated with a particular string—giving rise to resolved indeterminacy (see also Bergen, Levy, and Goodman 2016; Potts et al. 2016; Monroe 2018 for related approaches). We illustrate how the distinction between these forms of indeterminacy can be implemented through an example involving vague adjectives. We then illustrate a minimalistic analysis of factivity in Section 3.7.

3.1 Distributions on discourse constructs

Before launching into a probabilistic analysis of clause-embedding predicates, we lay out the core principles of PDS—in particular, how it supports dynamic analyses of discourse phenomena and the lexical semantics of expressions which trigger them, how it incorporates probabilistic knowledge into these analyses, and how it may be used to generate predictions about people’s behavior in the context of particular linguistic inference tasks.

In PDS, an ongoing discourse is represented by a map from an input state to a probability distribution over output states. States are tuples of *resolved parameters*; these include, e.g., the common ground (or *context set*; Stalnaker 1978) and the QUD, as well as other conversationally relevant features of discourse (e.g., possible antecedents for elliptical constructions). One can view the state as akin to the context state of Farkas and Bruce (2010), which includes representations such as a QUD stack and a common ground, among other constructs; PDS states are less constrained than Farkas and Bruce states in principle, in the sense that they may define arbitrary discourse constructs.

We notate the types of these state tuples ‘ σ ’, ‘ σ' ’, etc., so that an ongoing discourse is of type $\sigma \rightarrow P\sigma'$, where $P\sigma'$ is the type of *probability distributions* over states of type σ' . In words: a discourse is a map from input states to probability distributions over output states. Generally, $P\alpha$ is the type of probability distributions over values of type α . When we have a term of type $P\alpha$, we sometimes refer to it as a *probabilistic program*.

What is a probabilistic program? In PDS, $P\alpha$ is a type inhabited by λ -terms, just like all other types are. But whereas the meaning of a λ -term that inhabits a function type, $\alpha \rightarrow \beta$, is a function which, given something of type α , computes something of type β , the meaning of a λ -term that inhabits the type $P\alpha$ is a computation of a random value of type α according to some probability distribution. How exactly this computation is carried out—whether, e.g., by analytically computing a closed-form representation of the density associated with an arbitrary value of type α , or by approximating this density by sampling values directly from the relevant distribution, or by yet some other method—is immaterial. The purely declarative language of λ -terms makes space to do one thing only: describe the sort of computation of type $P\alpha$ which must be carried out in order to compute a random value of type α . Describing such a computation answers questions such as, “What information is this random value conditional on?”, “Does it depend somehow on

a value computed from a particular distribution?”, “Are there values it is blocked from taking on?”. That is, it provides a complete mathematical description of any random variable of interest in terms of the dependencies it enters into in virtue of the control flow of the program. By remaining indifferent about implementation details, we are afforded a very useful abstraction that puts probabilistic programs on the same plane as other typed λ -terms. Moreover, this abstraction allows such programs to be integrated into the ambient calculus seamlessly, without changing its meaning. Meanwhile, the monadic interface endows probabilistic programs with the technology required to compose them together while preserving their internal structure.

To perform updates of various types to an ongoing discourse, we use two operators—*bind* and *return*—which allow probability distributions to be sampled (bind) and ordinary, non-probabilistic values to be returned as *degenerate* distributions over those values (return).⁹ Both bind and return are indispensable if one’s aim is to characterize arbitrary probability distributions *compositionally*—i.e., to construct programs any of whose syntactic parts has a fixed (e.g., model-theoretic) interpretation. An interface for characterizing programs compositionally is indeed *necessary* if we aim to also characterize natural language meanings compositionally, where such meanings *depend* on the meanings of probabilistic programs. If we wish to compose linguistic meanings using functional application (for example), then our compositional regime already has its own interface with an attendant structure: we must view both the function and the argument—which may depend on a probability distribution—as objects that inhabit the right kind of space. The monadic treatment renders both interfaces fully compatible, allowing us to speak about functional application and probabilistic uncertainty in the same breath. As a result, we can develop semantic theories of natural language without compromising on either the semantic interface or the probabilistic interface at any point.

To illustrate, suppose we have some categorical distribution $\text{mammal} : Pe$ on mammals (where e is the type of entities). We can represent a distribution on mammals’ mothers as:

$$x \sim \text{mammal} \\ \boxed{\text{mother}(x)}$$

Here, $\text{mother} : e \rightarrow e$ maps an entity to its mother. Meanwhile, x is sampled from mammal using bind, and $\text{mother}(x)$ is then *returned*, as notated by the orange box. The idea is that this distribution assigns to any given entity y the probability $\sum_{x \in \text{mother}^{-1}(y)} p_{\text{mammal}}(x)$ (where $\text{mother}^{-1} : e \rightarrow e \rightarrow t$ is the inverse of the mother function and $p_{\text{mammal}}(x)$ is the probability that mammal assigns to x).

In a similar fashion, given two discourses $D_1 : \sigma \rightarrow P\sigma'$ and $D_2 : \sigma' \rightarrow P\sigma''$, we can invoke bind to sequence D_1 with D_2 and obtain a new discourse $D_3 : \sigma \rightarrow P\sigma''$:

$$D_3 = \lambda s. \left(\begin{array}{c} s' \sim D_1(s) \\ D_2(s') \end{array} \right)$$

Somewhat metaphorically: given a starting state s , we sample an output state from $D_1(s)$ and apply D_2 to this output state. Moreover, generalizing the previous example, given an input state

⁹Importantly, these operators conform to the monad laws. See Grove and Bernardy 2023; Grove and White 2024 for why this fact is useful.

s , the probability $p_{D_3(s)}(s'')$ that $D_3(s)$ assigns to an output state s'' is computed as:

$$p_{D_3(s)}(s'') = \sum_{s'} p_{D_2(s')}(s'') * p_{D_1(s)}(s')$$

In words, sequencing two maps from input states to probability distributions over output states involves doing pointwise multiplication of the probabilities assigned to intermediate states by the first map and those assigned to an output state by the second map and then summing the result.

Finally, we represent the common ground as a probability distribution over indices representing information about possible worlds, along with certain linguistic parameters (which, for current purposes, we keep fairly open ended); we refer to the type of indices as ' ι ', so that common grounds themselves are distributions of type $P\iota$. Given some index $i : \iota$, we refer to the “world” of i as ' w_i ' and to what we call the “context” of i as ' c_i '; contexts are what we use to encode useful linguistic information. The idea is that worlds w determine, e.g., how tall a particular individual is, while contexts c determine, e.g., the vague threshold of height past which one's height is considered tall. Thus while a world-sensitive function such as $\text{height}(w_i) : e \rightarrow r$ intuitively says something about how tall an entity is at an index i , the adjective's threshold $d_{\text{tall}}(c_i) : r$ only provides information about an aspect of the lexical semantics of *tall* at that index, viz., what is required of an entity to count as tall.

3.2 Making an assertion

Similar to discourses, the meanings of expressions are both probabilistic and dynamic. Accordingly, we represent them as functions of type $\sigma \rightarrow P(\alpha \times \sigma')$: given an input state, the meaning of an expression produces a probability distribution over *pairs* of ordinary meanings of type α and possible output states. Furthermore, given a sentence whose probabilistic dynamic meaning ϕ is of type $\sigma \rightarrow P((\iota \rightarrow t) \times \sigma')$, we can represent an *assertion* of that sentence as a discourse which updates the common ground. Specifically, we have a function `assert` with the following type:

$$\text{assert} : (\sigma \rightarrow P((\iota \rightarrow t) \times \sigma')) \rightarrow \sigma \rightarrow P\sigma'$$

Given such a ϕ , $\text{assert}(\phi)$ is a discourse of type $\sigma \rightarrow P\sigma'$ representing an assertion of ϕ . For any input state $s : \sigma$, $\text{assert}(\phi)(s)$ samples a proposition p together with an output state s' from ϕ and simply updates the common ground of s' with p ; in particular, it gives back a new state s'' just like s' , except that the common ground of s'' ($\text{CG}(s'')$) is just like the common ground of s' ($\text{CG}(s')$), but updated with p . Ultimately, assertions modify an ongoing discourse so that its probability distribution over output states features common grounds in which the proposition returned by ϕ has been observed to hold true (see Grove and White 2024 for further details).

3.3 Asking a question

We follow a categorial tradition by analyzing questions as denoting—given an index—sets of true short answer meanings (Hausser and Zaefferer 1978; Hausser 1983; Xiang 2021; cf. Karttunen 1977; Groenendijk and Stokhof 1984). For simplicity (though not by necessity), we assume that all questions are degree questions: given an index, they denote sets of degrees of type r (real numbers). Questions therefore have probabilistic dynamic meanings of type $\sigma \rightarrow P((r \rightarrow \iota \rightarrow t) \times \sigma')$.

Asking a question is a matter of pushing a question meaning onto the QUD stack (Roberts 2012; Farkas and Bruce 2010). Reflecting this, there is a function *ask* with the following type:

$$\text{ask} : (\sigma \rightarrow \text{P}((r \rightarrow \iota \rightarrow t) \times (\sigma'_1 \times \delta \times \sigma'_2))) \rightarrow \sigma \rightarrow \text{P}(\sigma'_1 \times ((r \rightarrow \iota \rightarrow t) \times \delta) \times \sigma'_2)$$

Given a probabilistic dynamic question meaning κ and an input state s , $\text{ask}(\kappa)(s)$ samples a pair of a question meaning $q : r \rightarrow \iota \rightarrow t$ and a state $s' : \sigma'_1 \times \delta \times \sigma'_2$ from $\kappa(s)$ and then updates the QUD stack of s' (something of type δ) by pushing q onto it (which we represent in terms of pairing). Thus it returns a new output state of type $\sigma'_1 \times ((r \rightarrow \iota \rightarrow t) \times \delta) \times \sigma'_2$.

3.4 Responding to a question

PDS also models responses to questions; at any point in an ongoing discourse in which the QUD stack is not empty, one can respond to the QUD at the top of the stack based on one's prior knowledge. Concretely, a given responder has some background knowledge $bg : \text{P}\sigma$ constituting a prior distribution over *starting* states. The responder uses this prior, in conjunction with the interim updates to the discourse, to derive a probability distribution over answers to the QUD. This answer distribution is gotten by retrieving the QUD of any given state s' —resulting in a distribution over QUDs—and then choosing a degree of which this QUD is true, given an index sampled from the common ground—resulting, finally, in a distribution over degrees. We further elaborate on these ideas in the next subsection.

3.5 Linking assumptions

In practice (e.g., in the setting of a formal experiment), an answer needs to be given using a particular testing instrument. We assume that a given testing instrument may be modeled by a family f of distributions representing the likelihood, which is then fixed by a collection Φ of nuisance parameters. Thus we may define a family of *response functions*, parametric in the testing instrument (i.e., likelihood function), each of which takes a distribution representing one's background knowledge, along with an ongoing discourse:

$$\text{respond}^{f_\Phi : r \rightarrow \text{P}\rho} : \text{P}\sigma \rightarrow (\sigma \rightarrow \text{P}\sigma') \rightarrow \text{P}\rho$$

For a fixed likelihood function f_Φ mapping any given degree answer onto a distribution over possible responses of type ρ (for some ρ), the response function takes a distribution representing background knowledge and a discourse to produce a response distribution. It does this by composing the discourse with background knowledge, as above, and then obtaining a degree answer to the current QUD, before applying the likelihood function f_Φ to this degree (see Section 4.1.1 for further details).

The testing instrument employed in our experiments, for example, is a slider scale that records responses on the unit interval $[0, 1]$. A suitable likelihood would therefore be a truncated normal distribution: $f(x, \Phi) = \mathcal{N}(x, \sigma) \upharpoonright [0, 1]$ (so that $f = \mathcal{N}$ and $\Phi = \sigma$). This likelihood—which we employ in our models of factivity—can be viewed as allowing some distribution of response errors, given the intended target response (i.e., the answer to the question).

3.6 Two kinds of uncertainty

We rely on the distinction encoded in PDS between uncertainty about the discourse state and uncertainty in the common ground to model the general behavior of resolved and unresolved indeterminacy, respectively. For illustration, take the sentence in (13).

(13) My hands are dry.

As Kennedy and McNally (2005) note, (13) contains an ambiguity brought on by the two senses of the adjective *dry*: on one reading, (13) is a claim about the quality of the speaker’s skin, while on another, it is a claim about the amount of liquid with which their skin is in contact. Kennedy and McNally argue that these two senses of *dry* associate it with *relative* (i.e., vague) and *absolute* standards of dryness, respectively. While relative standards are theorized as shifting from one context of use of the adjective to another, absolute standards are theorized (a) to be lexically fixed and, generally, (b) as coinciding with an *endpoint* on the scale that the adjective lexically encodes. Kennedy and McNally show that the two relevant senses can be brought out by modifiers which are typically compatible with only one or the other type of standard.

- (14) a. My hands are very dry.
b. My hands are partially dry.

While (14a) has “skin quality” sense (the relative sense), (14b) has the “contact with liquid” sense (the absolute sense).

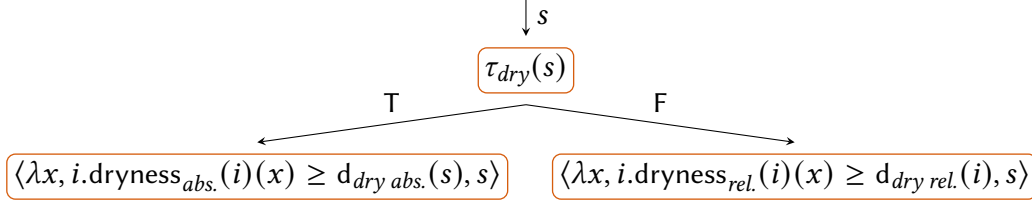
We can model this ambiguity by interpreting *dry* as the probabilistic semantic value in (15)—something of type $\sigma \rightarrow P((e \rightarrow \iota \rightarrow t) \times \sigma)$.

$$(15) \quad \lambda s. \begin{cases} \langle \lambda x, i. \text{dryness}_{abs.}(w_i)(x) \geq d_{dry\ abs.}(s), s \rangle & \tau_{dry}(s) \\ \langle \lambda x, i. \text{dryness}_{rel.}(w_i)(x) \geq d_{dry\ rel.}(c_i), s \rangle & \neg \tau_{dry}(s) \end{cases}$$

Given some input state s , *dry* returns one of two possible senses—either the relative sense or the absolute sense—as according to a truth value $\tau_{dry}(s)$ that depends on s . In case $\tau_{dry}(s) = 1$, the adjective takes on the absolute sense and is true of an entity at an index i of the common ground just in case the entity’s dryness at i exceeds some globally determined threshold $d_{dry\ abs.}(s)$ which depends on the state. If it takes on the relative sense, it is true of an entity at i if the entity’s dryness at i exceeds a threshold which *also* depends on i . The idea is that there is unresolved indeterminacy about the threshold of the adjective when this adjective is understood in its relative sense, and thus as vague; such uncertainty is usefully viewed as an aspect of the adjective’s core meaning, since it is associated with variability in how this meaning, when predicated of a particular entity, is true or false at different indices of the common ground. When the adjective is understood in its absolute sense, the uncertainty about the threshold is instead *resolved* uncertainty. Thus when its meaning is predicated of an entity, it may be true or false at an index of the common ground depending *only* on the properties of this entity—*not* variation in the degree threshold. Note that we consider the question of how to properly analyze the relative/absolute distinction to be open, in principle; the foregoing discussion provides one particular analysis of it.

In the meantime, we also introduce a *directed acyclic graph* (DAG) notation that allows such meanings to be rendered more perspicuously.¹⁰ We can give an alternative representation of (15) as a DAG, as in (16).

(16)



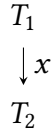
The idea is that each node of the DAG in (16), except for the root, represents a probability distribution (here, a degenerate distribution formed from a use of return); meanwhile, an edge going from a parent node to its child node represents how one probability distribution (the child) depends on the value sampled from another (the parent). An edge is labeled with the value which is sampled in order to generate the distribution represented by its target node. For example, the value $\tau_{dry}(s)$ may equal either T or F; when it is T, one can obtain the resulting probability distribution by looking to the child on the left, and when it is F, one can look to the child on the right.

In general, we will write either a literal value—e.g., T or F—or a variable name—e.g., when it is impractical to specify every possible value as a literal. Also note that the root of the DAG in (16) is unlabeled; this represents the dependence of the entire probability distribution on the input state s , which in (15) is encoded as a λ -abstraction.¹¹

One convenient property of the DAG notation is the way it represents binding one probabilistic program to another. For example, given a DAG T_1 and a DAG



if T_1 represents a distribution over things of type α and the variable x is of type α , we may form a third DAG



—understood to represent the result of concatenating a copy of the second DAG to *each* terminal node of T_1 . The resulting DAG represents sampling x from T_1 and then using x to compute the probability distribution T_2 , thus registering a new dependency.

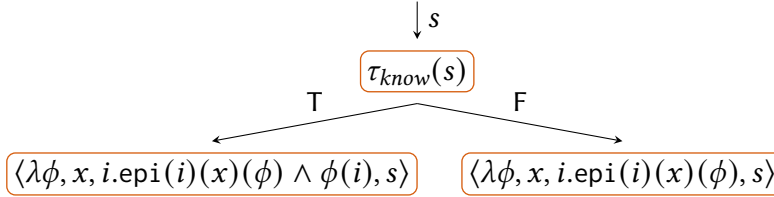
¹⁰This notation renders probabilistic meanings into a format somewhat resembling a Bayesian network (Pearl 1985). Indeed, the current notation may be understood as a Bayesian network which has each of its edges labeled by the variable sampled along that edge.

¹¹We may make sense of the convention of labeling branches with literal values by assuming that each such branch out of a node represents the program gotten by *conditioning* on the statement that the variable sampled from the node has the indicated value, and then inducing a uniform distribution over the resulting conditional programs.

3.7 The meaning of factivity

According to the Fundamental Discreteness Hypothesis, clause-embedding predicates entail the complement clauses they select with some probability in $[0, 1]$.¹² For example, one can represent the meaning of *know* as in (17), i.e., as a DAG.

(17)

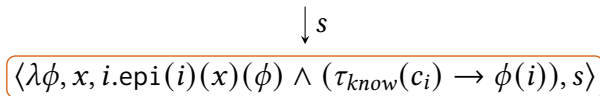


Here τ_{know} —analogous to τ_{dry} above—determines, based on the state s , whether to instantiate the meaning of *know* with a factive or a non-factive meaning. If $\tau(s)$ is T, *know* (a) ensures that the agent x is in the appropriate epistemic relation with the proposition ϕ denoted by the complement clause (whatever that may be), and (b) ensures that ϕ also holds at the index of evaluation. If $\tau(s)$ is F, the predicate only imposes the former requirement.

An analogous scheme holds for all clause-embedding predicates: those predicates which are always factive will have a meaning entailing their complement clause with probability 1, and those which are never factive will have a meaning which does not entail it with probability 1. Moreover, such variation among clause-embedding predicates is ultimately determined by the distribution over starting states for a given discourse. This prior distribution, we assume, encodes one’s core lexical knowledge, as it stands before any utterance is made, as well as other kinds of knowledge, e.g., about what the starting QUD is likely to be.

Importantly, the lexical entry in (17) renders *know* semantically ambiguous. Ambiguity, on our account, is a potential type of resolved indeterminacy, but not of unresolved indeterminacy; ambiguities are resolved on particular occasions of interpretation. The fact that an ambiguity is at stake is apparent in the shape of the DAG: the parent node labeled $\tau_{know}(s)$ has two children, each labeled by a possible meaning for *know*. An alternative lexical entry rendering this uncertainty as instead *unresolved* is the one in (18).

(18)



According to (18), whether or not the complement clause of the predicate is entailed depends on the index of evaluation: τ_{know} —which now has a different type!—determines a truth value based on *this index*, and, if this truth value is T, the meaning of the complement is also required to be T.

It is important to point out that the kind of characterization of factivity is *ad hoc*, in the sense that it doesn’t provide a characterization of the general *factors* distinguishing different predicates’ interpretations. But at the same time, we may say exactly how such factors ought to be encoded

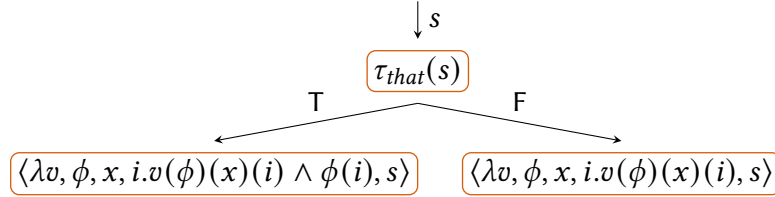
¹²We do not distinguish between factivity and veridicality for current purposes. This choice bears a resemblance to the general approach in Simons 2007.

in our system: they pertain to the shape of the state s which is read in by either of the above possible meaning representations for *know*, and in particular (a) the value s sets for $\tau_{\text{know}}(s)$ in (17), or alternatively (b) the value s sets for the common ground and the probability distribution over indices that this value encodes, which in turn determines the probability distribution over values for $\tau_{\text{know}}(c_i)$ in (18).

This approach is ideal for current purposes, as it allows us to characterize the behavior of the predicates we investigate at a level of abstraction that remains uncommitted to any particular view about the factors controlling factivity, but which may be used to circumscribe more detailed accounts. In particular, we wish to determine whether the gradience exhibited by clause-embedding predicates is a manifestation of unresolved indeterminacy (supporting the Fundamental Gradience Hypothesis) or resolved indeterminacy (supporting the Fundamental Discreteness Hypothesis). Thus we introduce just enough detail here to distinguish these hypotheses without also determining what causal factors might lead to one or the other being true.

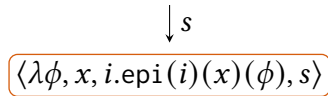
Note that alternatively, we could have described factivity as arising from a source external to the predicate; for example, by assuming that it is encoded by a complementizer (Kiparsky and Kiparsky 1970 *et seq.*).

(19) $\llbracket \text{that} \rrbracket =$



Given the alternative in (19), we need not assume that clause-embedding predicates, such as *know*, themselves give rise to factive interpretations.¹³

(20) $\llbracket \text{know} \rrbracket =$



What this sort of approach requires, in turn, is that contexts provide information about which predicate a given complementizer co-occurs with, which must be available in order for the probability of projection to be modulated by predicate type (see Gordon and Chafetz 1990; Trueswell, Tanenhaus, and Kello 1993; MacDonald, Pearlmutter, and Seidenberg 1994; Garnsey et al. 1997; Altmann and Kamide 1999, *i.a.*). If the context makes such information available, then this implementation can yield the same range of statistical models as one which assumes that factivity is driven principally by the lexical entries for individual predicates.

¹³The implementation according to which the semantic value of the complementizer operates on the semantic value of the predicate is analogous to some neo-Davidsonian approaches to the semantics of propositional attitude verbs and complementizers (see Kratzer 2006; Moulton 2009; Bogal-Allbritten 2016; White and Rawlins 2018a) and may be adapted to other such approaches that regard the embedded clause as an intersective modifier of eventualities (Elliott 2016; Elliott 2020).

4 Modeling

We now turn to the analyses of factivity and world knowledge which are possible under the framework described in Section 3 (see, in particular Sections 4.2 and 4.3). To investigate these analyses, we implement Bayesian models in the Stan programming language (Stan Development Team 2024) via the CmdStanR interface (Gabry and Češnovar 2023). We fit these models using Degen and Tonhauser’s (2021) experimental data and then compare them in terms of their expected log pointwise predictive densities (ELPDs) computed under the widely applicable information criterion (WAIC; Watanabe 2013; Gelman, Hwang, and Vehtari 2014), as implemented in R’s `loo` package (Vehtari et al. 2023). This measure quantifies how well each model fits the data, while also penalizing each model for how complex it is (i.e., the effective number of parameters it uses to fit the data).

In Section 4.1, we formalize our assumptions about the link between higher-order probabilistic programs of the kind described in Section 3 and participants’ response behavior. In Section 4.2, we describe and compare the two models which we fit to Degen and Tonhauser’s norming experiment data: a gradient model and a discrete model. We then describe each of the models of factivity possible under our framework in Section 4.3 before reporting our comparisons of these models, given Degen and Tonhauser’s projection experiment data, in Section 4.4.

4.1 Linking to response behavior

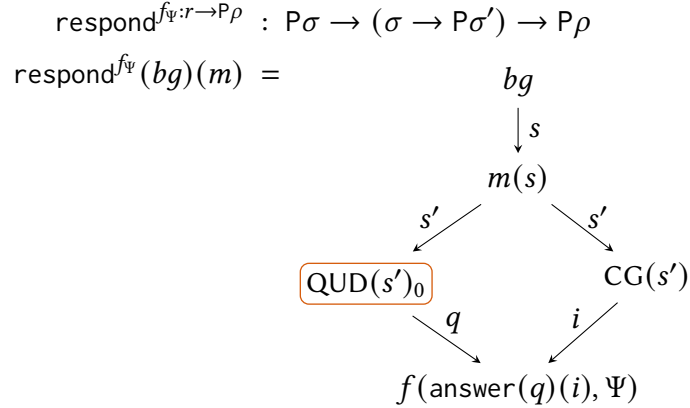
To connect the probabilistic programs associated with sentences to actual data, we need linking assumptions that relate these programs to the inference judgments that experimental participants report on a slider scale going between *no* and *yes* (cf. Jasbi, Waldon, and Degen 2019; Waldon and Degen 2020).

We assume that a slider scale can take on values in the closed unit interval $[0, 1]$. Specifying linking assumptions therefore requires saying how probabilistic programs may be used to compute distributions on this interval.

4.1.1 Response models in the abstract

Here we go into more detail about the response functions introduced in Section 3.5. Recall that these are of type $P\sigma \rightarrow (\sigma \rightarrow P\sigma') \rightarrow P\rho$: they take a probabilistic program *bg* of type $P\sigma$ representing a prior distribution over discourse states, along with an ongoing discourse (*m*, of

type $\sigma \rightarrow P\sigma'$), and associate them with a response distribution determined by the current QUD:

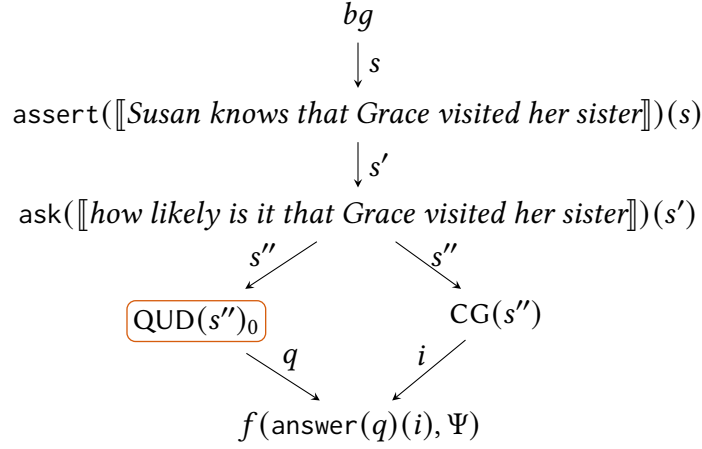


Going from top to bottom, it can be seen that a response is computed by: (a) sampling a state s from the prior distribution bg ; (b) applying the discourse m to s to obtain a distribution over outputs states; (c) sampling a new state s' from this distribution and grabbing the QUD from s' (i.e., along the left branch, where $\text{QUD}(s')_0$ is the top element of the QUD stack), as well as an index from the common ground of s' (along the right branch); and finally (d) computing an answer using the QUD and the index, which is fed to the response instrument. Note that the single terminal node of this DAG has two parent nodes—given by the QUD (bound to q) and an index of the common ground (bound to i)—in order to encode the flow of information from both of these sources. Further, since we assume that questions denote sets of degree answers, and are thus of type $r \rightarrow \iota \rightarrow t$, *answer* is easily defined to take the maximum degree in a given set, at a given index:

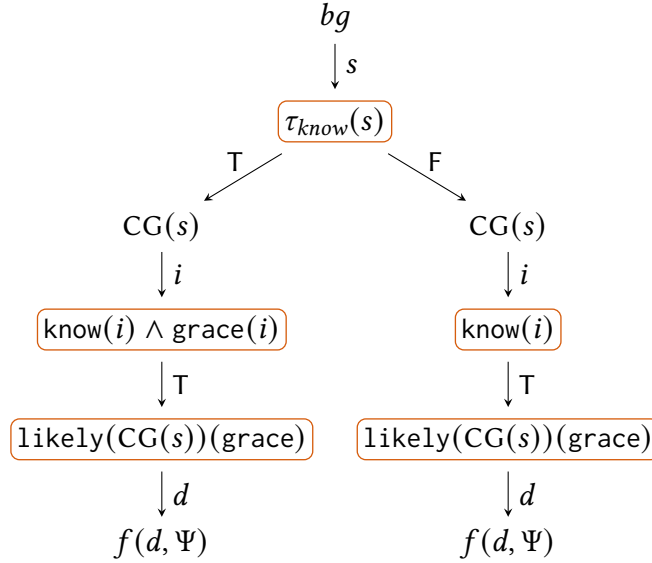
$$\text{answer}(q)(i) = \max(\lambda d. q(d)(i))$$

This model of response behavior assumes that participants compute the probability of an answer to a question prompt by determining whether or not the answer is true at a random index of the common ground, weighted by how likely that index is, given the prior bg updated with the discourse m . They then attempt to respond with this answer, i.e., by mapping the chosen degree d onto a physical response of type ρ (whatever that may be). But due to factors independent of the process by which d is computed (e.g., inaccuracies in their ability to perfectly target the response corresponding to d on the response instrument), they produce an actual response according to the probability distribution $f(d, \Psi)$ for some fixed family of distributions $f : r \times p_1 \times \dots \times p_n \rightarrow Pr$ and vector $\Psi : p_1 \times \dots \times p_n$ of nuisance parameters.

To give a schematic example, let's assume some prior bg of type $P\sigma$. Further, let's assume that the response instrument is a slider scale and can thus be encoded by the type r of real numbers, which we take to range in the interval $[0, 1]$. Then the following program of type Pr characterizes the distribution of slider responses in this interval, assuming $f_\Psi : r \rightarrow Pr$, where a response reflects a judgment about the likelihood that *Grace visited her sister* is true, given the information *Susan knows that Grace visited her sister*:



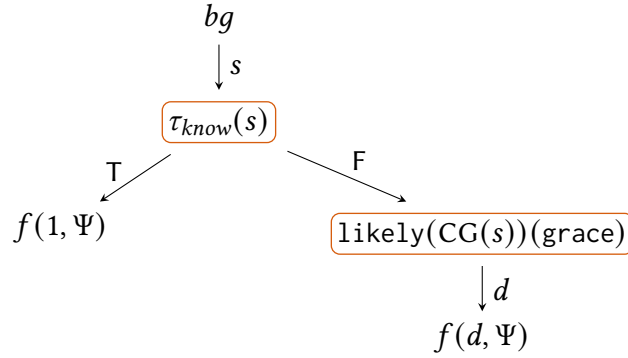
Suitable meanings for expressions like *likely*, along with bit of algebraic reasoning (the details of both of which we hide here for readability, but see Grove and White 2024) allows one to rewrite this program as the following one, assuming the *resolved* meaning for *know* of (17):



We employ a few abbreviations here to save space. $\text{grace} : \iota \rightarrow t$ abbreviates the meaning of the complement clause *Grace visited her sister*, while $\text{know} : \iota \rightarrow t$ abbreviates the epistemic component of the meaning of *know*, applied to the subject and the complement clause: $\lambda i.\text{epi}(i)(s)(\text{grace})$. Meanwhile, we assume the existence of a function

$$\text{likely} : P\iota \rightarrow (\iota \rightarrow t) \rightarrow r$$

such that $\text{likely}(cg)(\phi)$ is the probability of the proposition ϕ , given the distribution over indices encoded by the common ground cg . If we make the simplifying assumption that $\text{know}(i)$ and $\text{grace}(i)$ are *independent* in any common ground associated with a state sampled from the prior distribution (i.e., if Grace visiting her sister and Susan believing that Grace visited her sister don't influence each other's probabilities, according to any such prior common ground), then this program may be simplified even further:



In particular, the left branch may be simplified regardless of independence, while the right branch requires that $Pr(\text{grace}(i) \mid \text{know}(i), i \sim \text{CG}(s)) = Pr(\text{grace}(i) \mid i \sim \text{CG}(s))$. In practice, as we illustrate in Section 4.2, this independence assumption amounts to a justification for the method of directly using prior probabilities associated with world knowledge (e.g., about whether or not Grace is likely to have visited her sister) extracted from Degen and Tonhauser’s norming experiment, in order to generate one of the components in the mixed effects models described in Section 4.3.

4.1.2 Examples of response models

This general set-up will remain invariant under the models considered in the rest of this section. What varies is the lexical entries we associate with clause-embedding predicates like *know*, along with the meanings of their complement clauses; crucially, whether they are assumed to represent factivity and world knowledge as resolved or unresolved sources of uncertainty.

The remaining challenge in implementing our proposals is finding a family of distributions over $[0, 1]$ —the set of values we use to encode bounded slider scales—that can flexibly capture a variety of possible response distributions. Finding such a family is challenging insofar as many of the families of bounded distributions commonly used in generalized linear mixed models, such as the beta or logit-normal distributions, only have support on the open interval $(0, 1)$; i.e., beta and logit-normal random variables never take on 0 or 1 values. Because of this, they cannot describe the full distribution of slider responses, and in particular, the endpoints of the scale. This fact is particularly problematic in the current context, where endpoint responses are meaningful by hypothesis: consider the fact that according to the probability distribution illustrated above, the answer which is fed to the likelihood function f_Ψ is 1 when *know* is understood as factive.¹⁴

We consider two approaches to this challenge. The first is *truncation*, whereby we model distributions on the closed unit interval by *truncating* a distribution on r . The second is *inflation*, whereby we model 0 and 1 responses as arising from a response process that mixes a distribution on $(0, 1)$ with a distribution on $\{0, 1\}$, thereby *inflating* the number of 0’s and 1’s relative to that expected under the distribution on $(0, 1)$. To implement the truncation approach, we assume f to be the family of truncated normal distributions $\mathcal{N}(\cdot) \upharpoonright [0, 1]$, with nuisance parameter $\Psi \equiv \sigma$. To implement the inflation approach, we assume f to be the family of ordered beta distributions

¹⁴A common approach used in regression analyses is to sidestep this issue by “nudging” 0 and 1 responses toward 0.5 by a small δ , thereby putting them into $(0, 1)$. This approach is itself problematic because it introduces an additional unnecessary research degree-of-freedom; moreover, it is especially problematic in the current context for reasons which we discuss in Appendix A.

$\text{OrdBeta}(\cdot)$, with nuisance parameters $\Psi \equiv \langle \phi, \mathbf{c} \rangle$ (Kubinec 2023).¹⁵ Ultimately, we find that our main results are insensitive to which approach we take, so we focus on models implementing the truncation approach here (see Appendix B and Appendix C.2 for discussions of the inflation models).

4.2 Models of the norming data

To construct our models, we use a *pipelined* approach:¹⁶ we first fit models of Degen and Tonhauser’s norming data (experiment 2a), in order to obtain posterior distributions for parameters associated with the forty pairs of complement clauses and facts (what we refer to as *contexts*).¹⁷ We then used these posterior distributions as prior distributions for the corresponding parameters in the four models of Degen and Tonhauser’s projection experiment data (experiment 2b) which are implied by the framework described in Section 3. We present these models in Section 4.3.

Recall that the prompts with which participants were presented in the norming task inquired about the likelihood of a fact related to world knowledge, given some other background fact. To model such inference judgments, we fit two kinds of models: (i) a *gradient* model, according to which inferences conditioned by world knowledge are driven by unresolved indeterminacy; and (ii) a *discrete* model, according to which such inferences are driven by resolved indeterminacy. It turns out that the gradient model of the norming data provides the better fit (see Section 4.2.3). We therefore use the gradient model to extract prior distributions for parameters of the models of factivity.

4.2.1 World knowledge as a gradient phenomenon

The gradient model of the norming data encodes uncertainty about world knowledge in terms of unresolved indeterminacy. We give it schematically as follows, again calling the prior distribution ‘*bg*’, and assuming—following the question prompt used in Degen and Tonhauser’s task—that the response is a function of an answer to the question *how likely is it that S*:

$$\begin{array}{rcl}
 \text{norming-gradient} & = & bg \\
 & & \downarrow s \\
 & & \llbracket S \rrbracket(s) \\
 & & \downarrow \langle \phi, s' \rangle \\
 & & \boxed{\text{likely}(\text{CG}(s'))(\phi)} \\
 & & \downarrow d \\
 & & f(d, \Psi)
 \end{array}$$

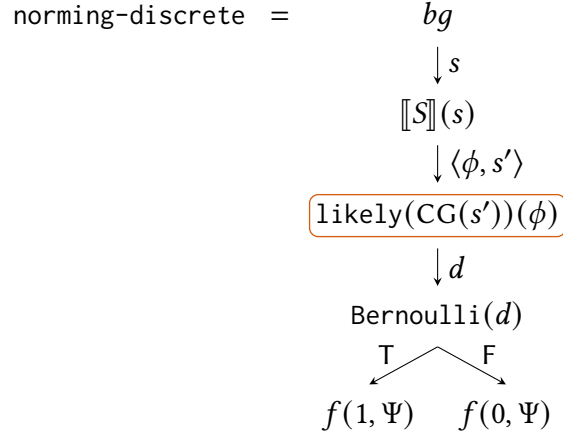
¹⁵See Appendix B for details on the ordered beta distribution.

¹⁶We provide additional formal details about the modeling pipeline in Appendix C.

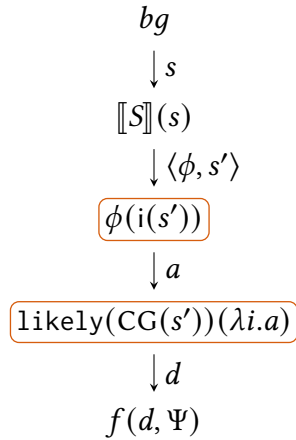
¹⁷We therefore overload the term ‘context’, but it should be clear on any given occasion of use in which sense we mean it.

4.2.2 World knowledge as a discrete phenomenon

The discrete model of the norming data encodes uncertainty about world knowledge in terms of resolved indeterminacy. We give it as follows:



The difference between this model and the gradient model is the continuation of the likelihood value with a Bernoulli program, which returns T with probability corresponding to that likelihood and F with a probability of one minus that likelihood. That is, the answer to the *how likely* question has been discretized, so that it is now 1 with probability determined by the original answer (and 0 otherwise). This model can alternatively be conceptualized as assuming that world knowledge is tied directly to the *discourse state*, as opposed to the common ground: instead of computing the truth of S relative to the common ground of the state s in order to determine its likelihood, such a program represents an index of evaluation as being obtained from s immediately ($i(s)$). It then computes a likelihood of either 1 or 0, depending on whether S is true or false at that index:



4.2.3 Comparison

The ELPDs for the two models of the norming data are shown in Figure 5 (left plot). It can be seen that the gradient model substantially outperforms the discrete model, indicating that responders represent their uncertainty about world knowledge gradiently.

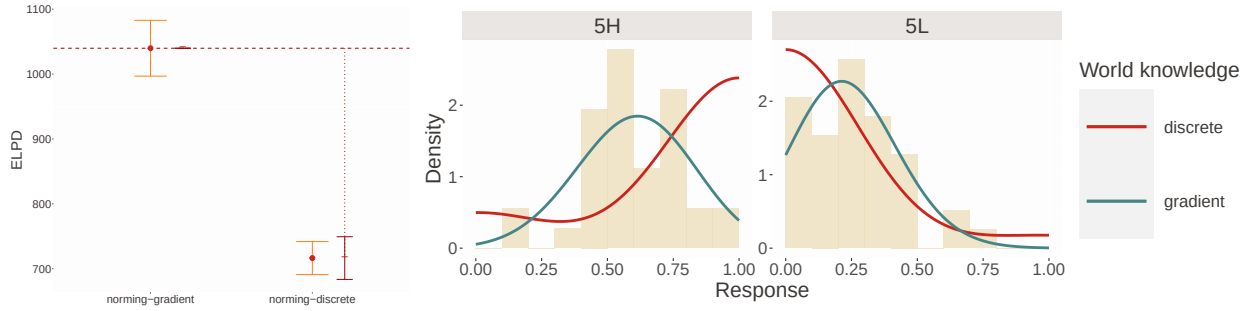


Figure 5: Left: ELPDs for the two models of the norming data. Dotted line indicates estimated difference between the norming-discrete model and the norming-gradient model. Error bars indicate standard errors. Right: posterior predictive distributions for both models for the item *Sophia got a tattoo*, given either the fact *Sophia is a hipster* (5H) or the fact *Sophia is a fashion model* (5L).

To give a sense of why the gradient model performs so much better, Figure 5 also shows the posterior predictive distributions for the two models for two items (right plot; see Figure 16 of Appendix D.2 for all items). These plots provide a visual indicator of how well each model fits the distribution of responses to the norming items: the closer a particular curve is to the shape of the histogram, the better the corresponding model fits the data. Thus, the gradient model fits the data better due to the fact that responses often cluster away from the endpoints of the scale, a pattern which the gradient model is adept at capturing. Meanwhile, the discrete model only allows the theoretical response (i.e., before any error is added) to occur at a scale endpoint. This limitation often causes the posterior predictive of the discrete model to have a bimodal distribution, as the plot for 5H illustrates.

4.3 Models of factivity

Now we specify our four models of factivity and prior world knowledge, which we fit to Degen and Tonhauser’s projection experiment data.¹⁸ Each of these models assumes that factivity is either discrete or gradient *and* that world knowledge is either discrete or gradient in their contributions to inference. Indeed, the models we present here make these assumptions wholesale, either regarding every predicate as making a discrete contribution or regarding every predicate as making a gradient contribution. Our current purpose is to compare these strong versions of the two hypotheses stated in Section 2. The possibility that predicates vary in whether the projective inferences they license are discrete or gradient is an interesting one, and we investigate it in ongoing work.

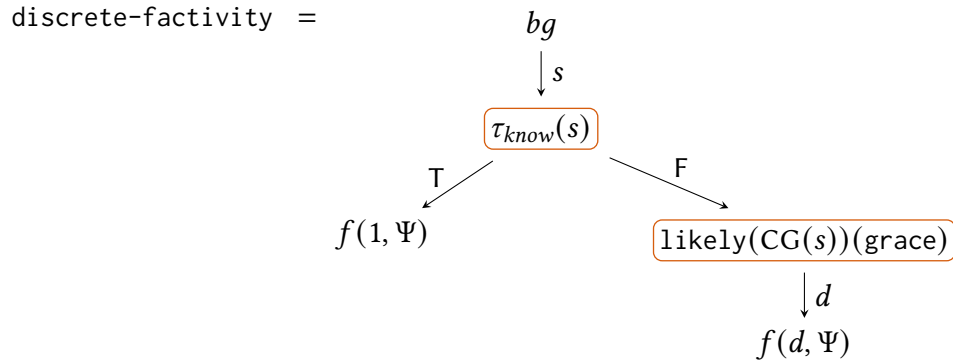
4.3.1 Factivity as a fundamentally discrete phenomenon

The first theory we consider regards sentences containing clause-embedding predicates as either triggering or not triggering factive inferences on particular occasions of use, depending on either (i) the interpretation of (a) the clause-embedding predicate, or (b) the surrounding structure; or (ii) some discourse structure, such as the common ground or question under discussion, whose properties are conditioned on pragmatic knowledge about the sentence. The uncertainty about

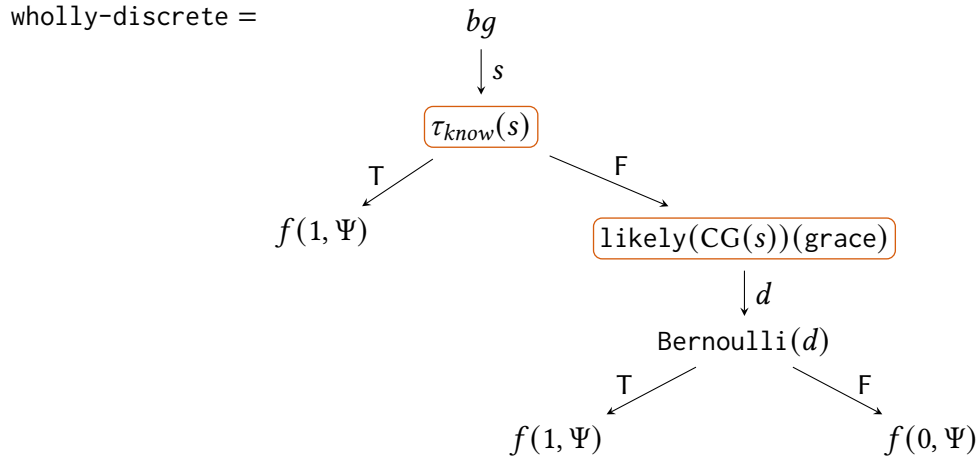
¹⁸See Appendix C for the full model specifications.

whether or not a given predicate’s complement clause projects is thus regarded as arising from resolved indeterminacy.

Assuming that factivity is a discrete phenomenon, the question remains how world knowledge affects inferences about the truth of an embedded clause. Here two possibilities arise. The first allows for uncertainty related to world knowledge to manifest as unresolved indeterminacy, which may, in turn, make individual judgments of truth uncertain. We refer to this theory as the *discrete-factivity* theory, emphasizing that it regards the contribution factivity makes as fundamentally discrete in nature. It gives rise to the model illustrated in Section 4.1.1, reproduced here, again for the predicate *know* and the complement clause *Grace visited her sister*:¹⁹



The second possibility is that both factivity and world knowledge are discrete in nature.



This *wholly-discrete* model effectively hypothesizes that no inferences display uncertainty on particular occasions of use.²⁰ Any gradience displayed in people’s measured inferences must therefore be due to task effects, as modeled by f_Ψ .

¹⁹Each of our models of factivity encodes the assumption that the QUD is a *how likely* question (e.g., *how likely is it that Grace visited her sister*). We use this choice of QUD type to yield degree questions, which we take to approximate the actual polar questions Degen and Tonhauser use in their experiments. See Grove and White 2025, which shows that collecting data under a task featuring a degree question tends to improve overall fit for models that appropriately account for the semantics of the question prompt.

²⁰We do not take anyone to endorse such a model; and indeed, the results from our fits to the norming data speak against it. We include it mainly because it is a logical possibility under our framework.

Note that the discrete-factivity model effectively provides a more complete version of the norming-gradient model described in Section 4.2, whereas the wholly-discrete model effectively completes the norming-discrete model. We therefore might have an *a priori* expectation that of the two, the discrete-factivity model should provide a better fit to the data, mirroring the superiority of the norming-gradient model to the norming-discrete model.

4.3.2 Factivity as a fundamentally gradient phenomenon

The theory which regards factivity as fundamentally gradient in nature does so by pushing what would otherwise be resolved indeterminacy about factivity onto the unresolved indeterminacy level. This theory also has two possible implementations that vary with respect to whether world knowledge is modeled as token-level indeterminate or type-level indeterminate. We refer to the version of the theory that regards both factivity and world knowledge as token-level indeterminate as the *wholly-gradient* version.

We obtain the corresponding model by modifying the discrete-factivity model so that it is effectively constructed from the lexical entry for *know* in (18), rather than the one in (17):

$$\begin{array}{c}
 \text{wholly-gradient} = \quad \quad \quad bg \\
 \quad \quad \quad \downarrow s \\
 \quad \quad \quad \boxed{\lambda i. \tau_{\text{know}}(i) \vee \text{grace}(i)} \\
 \quad \quad \quad \downarrow \phi \\
 \quad \quad \quad \boxed{\text{likely}(\text{CG}(s))(\phi)} \\
 \quad \quad \quad \downarrow d \\
 \quad \quad \quad f(d, \Psi)
 \end{array}$$

We refer to the alternative version of the model that regards world knowledge as discrete and factivity as gradient as the *discrete-world* version. As with the wholly-discrete model, we do not take anyone's theory of factivity to be consistent with the discrete-world model, in particular, but we include it, since it is a possibility implied by our framework. Indeed, this model is the mirror image of the discrete-factivity model:

$$\begin{array}{c}
 \text{discrete-world} = \quad \quad \quad bg \\
 \quad \quad \quad \downarrow s \\
 \quad \quad \quad \boxed{\text{grace}(i(s))} \\
 \quad \quad \swarrow \text{T} \quad \quad \searrow \text{F} \\
 \quad \quad f(1, \Psi) \quad \quad \boxed{\text{likely}(\text{CG}(s))(\lambda i. \tau_{\text{know}}(i))} \\
 \quad \quad \quad \quad \quad \downarrow d \\
 \quad \quad \quad \quad \quad f(d, \Psi)
 \end{array}$$

Among the four models we consider, we take the wholly-gradient model to come the closest to implementing the Gradient Projection Principle (Tonhauser, Beaver, and Degen 2018). Note

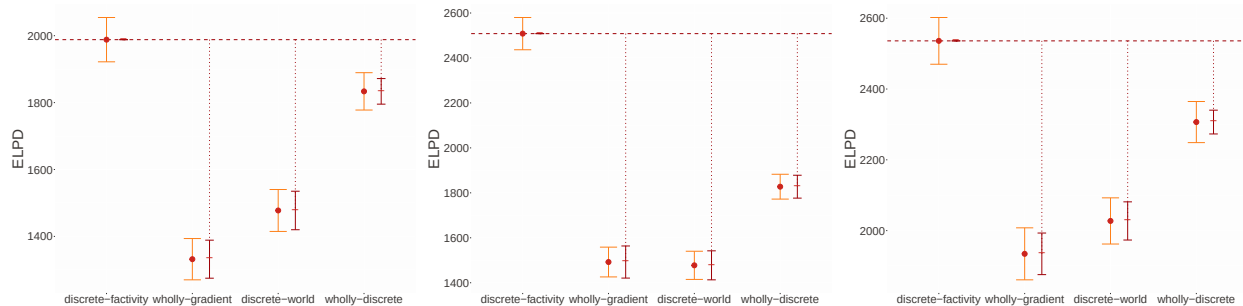


Figure 6: Left: ELPDs for the four models. Center: ELPDs for the four models with an anti-veridicality component added for each predicate. Right: ELPDs for the four model evaluations on our replication experiment data. Dotted lines indicate estimated differences between each model and the discrete-factivity model. Error bars indicate standard errors.

that this model effectively provides an alternative completion of the norming-gradient model described in Section 4.2; meanwhile, the discrete-world model provides an alternative completion of the norming-discrete model.

4.4 Comparisons

Figure 6 (left plot) provides ELPDs estimated for the four models, based on log-likelihoods computed from Degen and Tonhauser’s experimental data. We observe that the discrete-factivity model captures the data the best, while the wholly-discrete model trails behind it; meanwhile, the wholly-gradient and discrete-world models—the two that assume factivity to be gradient—perform the worst.

Thus, we have preliminary evidence that the best model of Degen and Tonhauser’s data treats factive presupposition projection as a discrete phenomenon and the inferences contributed by world knowledge as gradient. Further, note that by simply modifying the discrete-factivity model so that it treats factivity as gradient, one goes from the *best*-performing model to the *worst*-performing one.

To give a sense of the performance of the models as assessed against the inference judgment data, the posterior predictive distributions for each model are plotted for six predicates, for all contexts combined, in Figure 7 (see Figure 17 of Appendix D.2 for all predicates). We observe here that the models that assume that either factivity or world knowledge is discrete are better able to capture dips in the frequency of responses in the middle of the scale than the wholly-gradient model. As one might expect, the wholly-gradient model predicts distributions that are much smoother than the models that assume some form of discreteness. This behavior is the main reason the wholly-gradient model fits the inference judgment data the worst of any of the models.

One reason the models assuming some amount of discreteness can capture the dips in frequency toward the middle of the scale is that they effectively model the distribution over inference judgments as a mixture of distributions: at least one with a mode at 1 and another with a mode determined by the structure of the particular model. In contrast, the wholly-gradient model

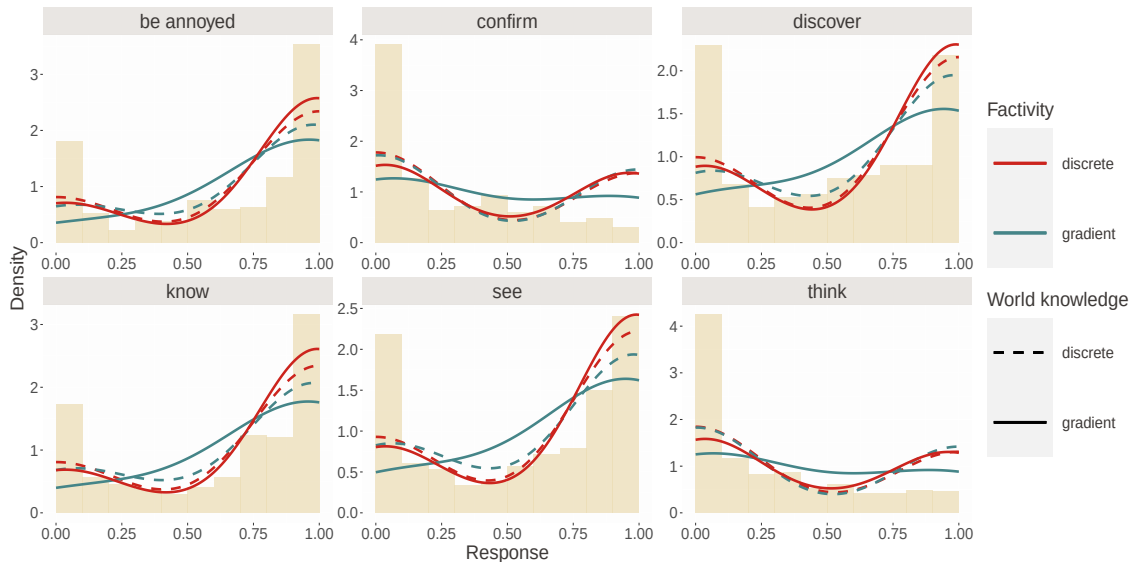


Figure 7: Posterior predictive distributions (with simulated participant intercepts) of all four models for six predicates from Degen and Tonhauser’s (2021) projection experiment, for all contexts combined. Empirical distributions are represented by density histograms of data from Degen and Tonhauser 2021.

only assumes a single distribution.²¹ This property of the models that assume some amount of discreteness makes them more complex than the wholly-gradient model—in the sense that they have more effective parameters—but as the pattern of ELPDs in Figure 6 shows, this additional complexity is offset by how much better they fit the data. (Recall that ELPD explicitly quantifies how well a model fits the data while penalizing model complexity, as measured by the effective number of parameters.)

Notably, none of the four models fits the data perfectly. For instance, the canonically non-projective predicates *think* and *pretend* have distributions which all four of our models appear to have difficulty capturing, at least by visual inspection of Figure 17. This difficulty appears to be due to an anti-veridicality inference associated with these predicates; i.e., an inference that the content of the complement clause is *not* true.²²

Since we consider denotations for predicates that vary only with respect to veridicality and non-veridicality—not anti-veridicality—no model captures the latter. To assume a three-way distinction among veridicality, non-veridicality, and anti-veridicality would effectively be to allow the discrete-factivity and wholly-discrete models to mix in another distribution with a mode at 0; meanwhile, it would effectively allow the wholly-gradient model an extra parameter to modulate the gradience associated with individual predicates.

²¹Multi-modality may still arise in the posterior predictive distributions for the wholly-gradient model; see, e.g., the posterior predictive distribution for *confirm* in Figure 7. Such multi-modality can only arise due to participant random effects. See Appendix C for formal details concerning these random effects.

²²These inferences likely arise from different sources; e.g., the lexical semantics of *pretend*, but a conversational implicature in the case of *think*. More generally, inferences of this kind may in principle arise for *any* predicate on its non-factive interpretation (assuming factivity is discrete), which may shed additional light on the mass of responses at or close to 0 in the empirical data across predicates.

To investigate the possibility that such a modification might affect the model ranking, we additionally fit models which add an anti-veridicality component for each predicate (see Appendix C.1.2). We find that, while the anti-veridicality component substantially improves the performance of the discrete-factivity model (as illustrated by the plots in Figure 18 of Appendix D) the only change in relative model performance is that the wholly-gradient model now slightly outperforms the discrete-world model, as shown in Figure 6 (center plot).

5 Evaluations

Strictly speaking, the comparisons we report in Section 4 are *post hoc*; and while the results are suggestive, we cannot draw firm conclusions from these model comparisons without a replication of Degen and Tonhauser’s experiment. In Section 5.1, we report such a replication, finding the same pattern of model comparison results: the models that assume that factivity is discrete reliably outperform the models that assume it is gradient. To ensure that these results are not somehow driven by the particular discourse contexts used in Degen and Tonhauser’s experiments, we collect two additional datasets that use the same paradigm, but that mask the contents of the embedded clause in two distinct ways. In these additional experiments, which we report in Section 5.2, we again find that the discrete-factivity models outperform all of the other models.

5.1 Experiment 1: held-out projection experiment data

5.1.1 Materials

Our materials and methods were identical to those of Degen and Tonhauser (2021).

5.1.2 Participants

We collected data from 300 participants using Amazon Mechanical Turk, paying each participant one dollar. Each participant was required to pass the qualification test described in White, Hacquard, and Lidz 2018, in order to ensure that they were a native speaker of American English. We removed data from two participants who claimed to have technical difficulties completing the experiment, and from ten more whose performance was more than two standard deviations below the mean on the six control items, leaving us with data from a total of 288 participants.

5.1.3 Results

Figure 8 shows the item means from our replication experiment plotted against Degen and Tonhauser’s original experiment, along with the means obtained by collapsing across contexts. We observe that there is strong agreement (Spearman’s $r = 0.98, p \leq 0.001$) between these means and those obtained from Degen and Tonhauser’s data.

5.1.4 Model fitting

To evaluate the four models using this data, we obtain, from each model, means μ_v and standard deviations σ_v of the marginal posterior distribution of the log-odds of projection for each predicate, as well as means μ_ω and standard deviations σ_ω of the marginal posterior distribution

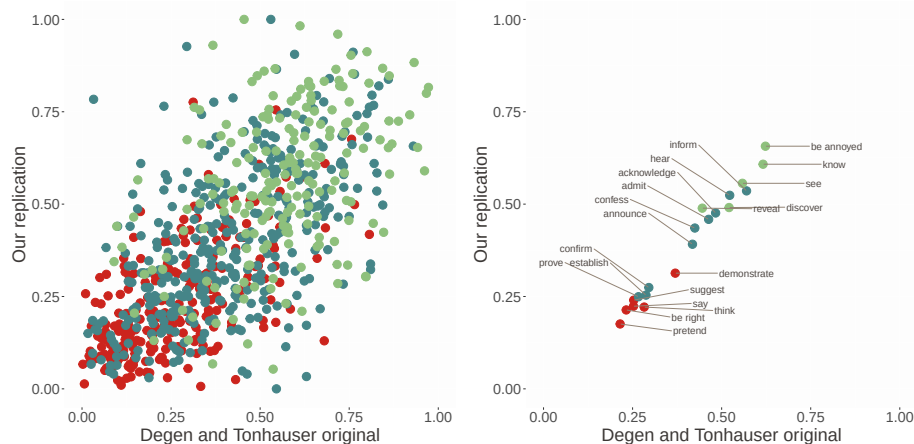


Figure 8: Degen and Tonhauser’s (2021) projection data versus our replication. Left: item means (Spearman’s $r = 0.68$, $p \leq 0.001$). Right: verb means (Spearman’s $r = 0.98$, $p \leq 0.001$). For both, “non-factive” verbs are in red, “optionally factive” verbs are in teal, and “canonically factive” verbs are in green.

of the log-odds certainty for each context. We then use normal distributions with these means and standard deviations as prior distributions for the corresponding parameters in the models constructed for the evaluations.²³

5.1.5 Model comparison

Figure 6 (right plot) provides ELPDs estimated for the four model evaluations, based on log-likelihoods computed from the data obtained in our replication experiment. The pattern of goodness-of-fit observed here is extremely close to that exhibited by the original model comparison on the left in Figure 6: the discrete-factivity model fares the best, followed by the wholly-discrete model. The wholly-gradient and discrete-world models fare the worst.

Thus, we have evidence that the differences in performance among these models that was reported in the previous section are quite robust, at least when assessed using data from Degen and Tonhauser’s experimental task. The next two experiments provide a test of the models in a somewhat different setting—one in which the uncertainty contributed by the linguistic contexts of the predicates of interest is sent to an extreme.

5.2 Experiments 2 and 3: non-contentful contexts

We now test the inferred posterior distributions over probabilities of projection on held-out data from two experiments in which the context of each predicate is stripped of the rich lexical content that partially governs the inferences produced in the original experiment. We obtain contexts for this evaluation in two ways. In our Experiment 2, we *bleach* each predicate’s complement clause so that it is just *a particular thing happened*. In the Experiment 3, we use a *templatic* complement clause of the form *X happened*.

²³See Appendix C.1.3 for further details concerning these models.

These manipulations serve two purposes. First, they allow us to assess the performance of the four models when the source of variance among inference judgments contributed by prior world knowledge is removed. Second, they put the predicates in environments in which knowledge about the context is minimal; as a result, they may produce greater uncertainty in people's inferences. This additional uncertainty could confer an *a priori* advantage to the wholly-gradient model, which considers all inferences, even those triggered by projective predicates, to be beset with some uncertainty. We thus consider these evaluations to be a stronger test of the discrete-factivity model's edge over the wholly-gradient model than the original experiments were.

5.2.1 Materials

To construct the bleached items, each of the twenty predicates investigated in Degen and Tonhauser's experiment was placed in a context in which its subject was one of the proper names from the original experiment, and in which its complement clause was just *a particular thing happened*. On each trial, participants were provided with a background context that was intended to make the prompt as natural as possible. The only thing that varied in this background context from one trial to the next was the name of the individual who makes the relevant utterance. Finally (taking that individual's name to be *P*), participants were prompted to answer the question *Is P certain that that thing happened?* on a sliding scale with *no* on the left and *yes* on the right. The following experimental trial, for example, involves the predicate *pretend*:

You are at a party. You walk into the kitchen and overhear Linda ask somebody else a question. Linda doesn't know you and wants to be secretive, so speaks in somewhat coded language.

Linda asks: "Did Tim pretend that a particular thing happened?"

Is Linda certain that that thing happened?

no yes

Next

In addition to the twenty bleached items, participants saw six control items which were constructed in order to somehow incorporate a bleached subordinate clause; for example, *Did Madison have a baby, despite the fact that a particular thing happened?*. All six control items had an intended response of 1.

5.2.2 Experiment 3: templatic items

To construct the templatic items, each of the same twenty predicates was placed in a context in which its subject was, again, a proper name from the original Degen and Tonhauser experiment, and in which its complement clause was *X happened*. A background context was provided on each trial, so that the prompt was natural. Background contexts, again, only differed from one another in the name of the individual who makes the relevant utterance. Given a trial on which

the individual P was the speaker, participants were prompted with the question *Is P certain that X happened?*, which they answered on a sliding scale with *no* on the left and *yes* on the right. The following example trial features the predicate *pretend*:

You are at a party. You walk into the kitchen and overhear William ask somebody else a question. The party is very noisy, and you only hear part of what is said. The part you don't hear is represented by the 'X'.

William asks: "Did Ray pretend that X happened?"

Is William certain that X happened?

noyes

Next

Participants again saw six control items which were constructed in order to incorporate a templatic subordinate clause; for example, *Did Madison have a baby, despite the fact that X happened?*.

5.2.3 Participants

For each experiment, we collect data from 50 new participants using Amazon Mechanical Turk, paying each participant one dollar. Each participant was again required to pass the qualification test described in White, Hacquard, and Lidz 2018, and any participant whose average score on the control items did not fall within two standard deviations below the mean of all participant's responses was excluded from the analysis. Using this criterion, three participants' data was excluded from Experiment 2, leaving 47 participants for analysis; and one participant was excluded from Experiment 3, leaving 49 participants for analysis.

5.2.4 Results

We observe in Figure 9 that the responses elicited by both the bleached (Spearman's $r = 0.97$, $p \leq 0.001$) and templatic (Spearman's $r = 0.87$, $p \leq 0.001$) items track the gradient knowledge about factivity that people deploy in the typical contentful setting extremely well. Not only is the same type of gradience observed among predicates when they are placed in bleached or templatic contexts, but the rankings among predicates are maintained almost entirely.²⁴ This finding furthermore suggests that it is safe to compare modeling results obtained from only bleached or templatic items to those obtained from contentful items—e.g., the clustering results of Kane, Gantt, and White 2022 discussed in Section 2—at least if those results pertain to the aggregate responses for individual predicates.

²⁴Notably, the range of average ratings for predicates in Experiment 3 is not as wide as exhibited in the previous experiments, with most falling between 0.5 and 0.75, suggesting that there may have been a great deal of uncertainty governing the inferences produced from this task.

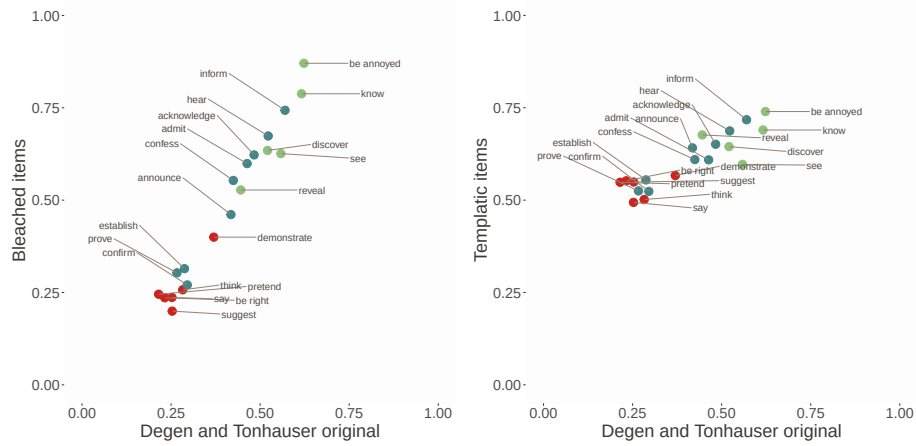


Figure 9: Degen and Tonhauser’s (2021) projection data versus data from contexts with minimal lexical content. Left: bleached data (Spearman’s $r = 0.97, p \leq 0.001$). Right: templatic data (Spearman’s $r = 0.87, p \leq 0.001$). For both, “non-factive” verbs are in red, “optionally factive” verbs are in teal, and “canonically factive” verbs are in green.

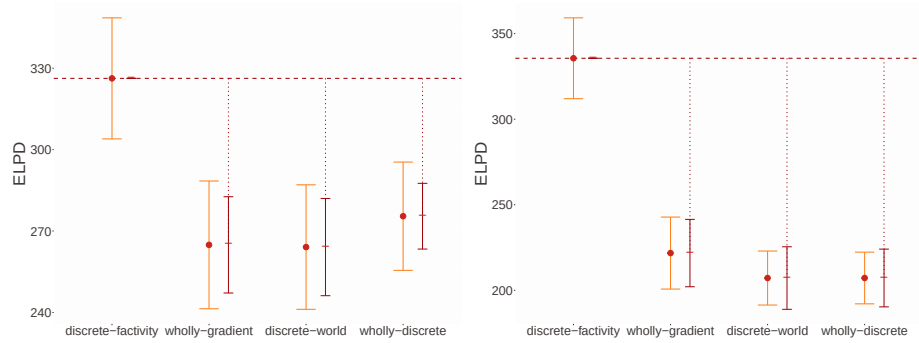


Figure 10: ELPDs for the four model evaluations on the bleached data (left) and the templatic data (right). Dotted lines indicate estimated differences between each model and the discrete-factivity model. Error bars indicate standard errors.

5.2.5 Model fitting

To evaluate the four models using both the bleached and templatic data, we use the same means μ_v and standard deviations σ_v of the marginal posterior distributions of the log-odds of projection that we used for the evaluations on the replication experiment data. As before, we use normal distributions with these means and standard deviations as prior distributions for the corresponding parameters in the models constructed for the evaluations. Then, in each evaluation, we infer a distribution over the parameters σ_ω and ω that regulate the certainty associated with either the bleached or the templatic context.²⁵

5.2.6 Model comparison

Figure 10 provides ELPDs for all four models evaluated on Experiments 2 and 3. We see here that the discrete-factivity model fares the best in both experiments, while the other three models fare comparably with each other. It is somewhat remarkable that the more fine-grained differences among models observable from both the original fits and the evaluation on the replication data do not appear to hold up under the current evaluation. For example, the wholly-discrete model no longer appears distinguished from the wholly-gradient and discrete-world models by its better performance. Rather, the discrete-factivity model seems to have a unique advantage.

This difference is especially notable for Experiment 3, given the somewhat squashed average responses seen across verbs in Figure 9. Despite the high amount of uncertainty which this task may have produced, such uncertainty seems to have been filtered through the discrete behavior of factive inferences. The uncertainty about such inferences appears to relate to the interpretation of a given predicate, rather than a gradient contribution which that predicate makes to an inference.

To give a sense of the differences among the model evaluations on the bleached and templatic data, Figures 19 and 20 of Appendix D.2 show the posterior predictive distributions of these evaluations for all predicates.

6 Consequences for theories of factivity

We have compared four models of the task reported in Degen and Tonhauser 2021, each fit to their experimental data. These models differ from one another along two axes. First, whether they consider the contribution of prior world knowledge to inferences about the truth of the clause embedded by a predicate to involve resolved or unresolved indeterminacy; and second, whether they consider the contribution of the relevant predicate itself to these inferences to involve resolved or unresolved indeterminacy.

Our initial comparison of the four models found that a model that associated world knowledge with unresolved indeterminacy and predicate knowledge as involving resolved indeterminacy best accounts for the distributions of judgments in Degen and Tonhauser’s experimental data. Moreover, follow-up evaluations of the four models confirmed this initial comparison: the same model best accounts for held-out data from a replication of Degen and Tonhauser’s experiment, for which distributions over the parameters of interest are extracted from the posteriors of the initial models (see Figure 6, right plot). The discrete-factivity model also best accounts for data from two tasks in which predicates are placed in contexts with minimal lexical content (see Figure 10).

Taken together, these results provide strong evidence that the gradience observed by White and Rawlins (2018b) and studied by Degen and Tonhauser is driven by *resolved* indeterminacy; i.e., it is associated with variation across experimental trials in the *type* of an expression and thus, whether or not it gives rise to a particular inference. This finding in turn suggests that clause-embedding predicates differ in the frequencies with which they trigger projective inferences, but the contribution a predicate makes on particular occasions of use (or interpretation) is *discrete*: either it produces the relevant inference, or it does not produce it at all. These results provide strong support for the Fundamental Discreteness Hypothesis, discussed in Section 2. Here we say

²⁵See Appendix C.1.4 for further details concerning these models.

a bit more about what we think they say for theories of factivity—and presupposition projection more generally.

In Section 2, we considered three ways in which the Fundamental Discreteness Hypothesis might be cashed out: (i) clause-embedding predicates may have multiple senses—at least one that is implicated in triggering projection and at least one that is not; (ii) clause-embedding predicates may occur in multiple structures—at least one that is implicated in triggering projection and at least one that is not; or (iii) clause-embedding predicates may be usable in contexts where the common ground—or some other construct, such as the question under discussion—entails the content of its embedded clause, as well as contexts where it does not (see Simons, Beaver, et al. 2017; Roberts and Simons 2024). Under the first two explanations, which we may describe as *conventionalist*, the resolved indeterminacy we observe is ultimately related to lexical or structural ambiguity; under the third explanation, which we may describe as *conversationalist*, this indeterminacy pertains to what kind of discourse contexts an expression tends to occur in. Importantly, modern accounts of presupposition projection tend to *blend* conventionalist and conversationalist strategies in explaining particular phenomena, with most accounts leaning more heavily in one direction or the other (see White 2019 for an overview). We illustrate this fact in Section 6.1 for one broadly conventionalist account of the projective inferences of emotive predicates (Djärv, Zehr, and Schwarz 2018). In Section 6.2, we briefly mention the conversationalist accounts of projection found in Simons, Beaver, et al. 2017 and Roberts and Simons 2024 which, taken together with our results, would require any indeterminacy about discourse constructs such as the QUD to be resolved in nature.

6.1 Conventionalist accounts

Consider an emotive predicate such as *be annoyed*. In addition to giving rise to projective inferences, such a predicate tends to produce inferences about the beliefs of the entity it is predicated of. For example, (21) tends to give rise to both the inferences in (22a) and in (22b).

(21) Was Jo annoyed that Mo left?

- (22) a. Jo believed that Mo left.
b. Mo left.

One species of conventionalist account attributes (22a) and (22b) to independent semantic properties of (21) (see Villalta 2008; Romero 2015; Uegaki and Sudo 2019, i.a.).

Another attributes both inferences to the same property, e.g., deriving one from the other by invoking additional pragmatic assumptions (Djärv, Zehr, and Schwarz 2018). For instance, adapting ideas put forth by Heim (1992, p. 212), (22b) might be derived from (22a) by invoking a defeasible assumption that Jo is credible. On this sort of account, (22a) and (22b) need not be triggered independently.

This second species of account can be extended to emotive factives as a class (Djärv, Zehr, and Schwarz 2018, p. 382), given that all emotive predicates tend to give rise to such belief inferences (see Kane, Gantt, and White 2022). Gradiance might then be a product of variability either in whether or not a belief inference is drawn or in whether or not an assumption of credibility is made.

6.2 Conversationalist accounts

In contrast, conversationalist approaches posit that knowledge of predicates like *be annoyed* relates those predicates to distributions over discourse-theoretic constructs, such as the common ground or the question under discussion. Such distributions are likely constrained by general pragmatic principles, which would, in turn, interact with conceptual knowledge associated with a given predicate to generate particular inferences (see Simons, Beaver, et al. 2017; Roberts and Simons 2024). Crucially, lexical semantic representations of presupposition are not the main driver of projective inferences under this sort of account—if they are implicated at all.

For instance, Simons, Beaver, et al. put forth a conversationalist account that relies on the notion of a QUD. On their account, whether or not the complement of a clause-embedding predicate projects varies according to prosodic and contextual factors conditioned by the QUD.²⁶ Specifically, Simons, Beaver, et al. argue that the projective inferences licensed at any point in a discourse are simply those which are backgrounded by the QUD, while the non-projective inferences are those which are made at-issue (see also Simons 2001; Simons 2007, i.a.). Simons, Beaver, et al. provide the example in (23) (their ex. 10) to illustrate this.

- (23) A: James just found out that Harry’s having a graduation party, and I can’t understand why he’s so upset about it.
 B: He didn’t find out that HARRY’s having a graduation party...
 ... he found out that HARRIET is having a graduation party, and HARRIET is his best friend.

Crucially, while the content of the complement of *find out* projects in other contexts, it does not appear to project in (23) due to the presence of focus on the embedded subject *Harry*. On the assumption that *find out* does not lexically trigger a projective inference, it ought not to trigger one in the context of (23) either, according to Simons, Beaver, et al., under the assumption that the QUD consists of alternatives according to which James found out that someone or other is having a graduation party.

Roberts and Simons (2024) extend this idea by arguing that “projectivity... is predictable from the detailed lexical semantics of the triggers, in combination with broad pragmatic principles” (p. 2), but crucially also that “projective readings of [change of state] predicates, of factives, and of predicates that trigger selectional restrictions, are not conventionally derived” (p. 29). Rather, they hold that these projective inferences are driven by “the nuanced and probabilistic knowledge that language users have about the functions, effects and contexts of use of linguistic forms” (p. 27), and they work through various specific examples.

²⁶Importantly, lexical knowledge also conditions QUD choice under standard dynamic accounts in the Karttunen-Heim tradition; it just does so indirectly. If, following Heim (1983), a sentence denotes a context-change potential, and its presuppositions are encoded as definedness conditions, then it will only be defined on context sets that satisfy its presuppositions. If (following, e.g., Roberts (2012)) a QUD is just a partition of the context set, the update denoted by a sentence is defined just in case that sentence’s presuppositions are true in every cell of the partition—i.e., if the QUD entails the sentence’s presuppositions. As a result, variability in a sentence’s definedness conditions may be seen to covary with QUDs insofar as the latter background the inferences triggered by those definedness conditions.

7 Conclusion

The results reported here can be understood as consistent with a view of factivity as a discrete property of an expression on a particular occasion of use. Sometimes, a predicate (depending on the predicate) gives rise to a projective inference, and sometimes it does not (again, depending on the predicate). It never partially projects. Indeed, this conclusion is in line with the view originally advocated by Kiparsky and Kiparsky (1970), Karttunen (1971), *inter alia*: according to this view, certain predicates can be understood to discretely trigger a presupposition that the clause they select is true.

But while we address the nature of gradience at the token level, our results do not speak to the importantly distinct question of whether there exist classes of factive predicates at the type level. Alone, our results are consistent with a view wherein there are no semantic classes of factive predicates. But in conjunction with the results of Kane, Gantt, and White (2022), discussed in Section 2, who do find the existence of clear factive classes of predicates (pace Degen and Tonhauser 2022), a clear picture emerges: in line with the traditional view, factivity displays discreteness at both the type and token levels. A major remaining question is exactly how this discreteness is derived: by conventional means, by conversational means, or (most likely) by some mixture of the two (Roberts and Simons 2024).

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A Beta distributions as likelihoods

Beta distributions cannot be used to model slider scales allowing endpoint responses. One common approach used in regression analyses sidesteps this issue by “nudging” 0 and 1 responses toward 0.5 by a small δ , thereby including them in $(0, 1)$. This approach is problematic because it introduces an unnecessary researcher degree of freedom—that of deciding by what amount δ to nudge the 0’s and 1’s inward—which can be used to manipulate model comparison metrics, and which we have no principled way of setting *a priori*.

First, note that WAIC relies crucially on the likelihood of the model. Second, note that the beta likelihood is highly sensitive to how much one nudges 0’s and 1’s inward, and that this sensitivity increases with the number of 0’s and 1’s present in the data. To see this, recall that the beta likelihood is:

$$\text{Beta}(\mathbf{x} \mid \alpha, \beta) = \frac{\prod_i x_i^{\alpha-1} (1 - x_i)^{\beta-1}}{B(\alpha, \beta)}$$

The log-likelihood is thus:

$$\mathcal{L}_{\text{Beta}}(\alpha, \beta \mid \mathbf{x}) = -\log B(\alpha, \beta) + \sum_i (\alpha - 1) \log x_i + (\beta - 1) \log(1 - x_i)$$

Now suppose that some of the $x_i \in \{0, 1\}$, and therefore that the beta distribution would be undefined, since we would be taking $\log 0$. Let’s say we nudge these x_i inward by $\delta < 0.01$ to

avoid this outcome. Then we would obtain the likelihood:

$$\begin{aligned}\mathcal{L}(\alpha, \beta \mid \mathbf{x}) = & -\log B(\alpha, \beta) + \sum_{i:x_i=1} (\alpha - 1) \log(1 - \delta) + (\beta - 1) \log \delta \\ & + \sum_{i:x_i=0} (\alpha - 1) \log \delta + (\beta - 1) \log(1 - \delta) \\ & + \sum_{i:x_i \notin \{0,1\}} (\alpha - 1) \log x_i + (\beta - 1) \log(1 - x_i)\end{aligned}$$

Thus when $x_i \in \{0, 1\}$, one term always contains a $\log(1 - \delta) \approx 0$ and another term always contains a $\log \delta$. The latter term is the problem: it is sent toward negative infinity as the nudging factor δ goes to 0.

As a result, manipulating δ in even minute increments can massively change the likelihood computation so that it pays more or less attention to 0's and 1's—in the limit, ignoring all responses $x_i \in (0, 1)$. Moreover, because the WAIC is computed in terms of the likelihood, manipulating δ can greatly affect model comparison results. Thus introducing such a nudge factor is problematic, given the methodological problems it poses; especially, when there are likelihoods which are actually *defined* for $x_i \in \{0, 1\}$. One such likelihood is given by the ordered beta distribution, which we look at next.

B Ordered beta distributions as likelihoods

Ordered betas models are a variant of the more general 0-1 inflated beta model (see Liu and Eugenio 2018 and references therein) that avoids some of the conceptual problems of the latter (Kubinec 2023). Intuitively, a 0-1 inflated beta likelihood models the response as a two-stage process: first, it decides whether to respond with 0 (which it does with probability $p_{\{0\}}$), with 1 (with probability $p_{\{1\}}$), or with some value in $(0, 1)$ (with probability $p_{(0,1)} = 1 - p_{\{0\}} - p_{\{1\}}$); second, if it decides to respond with some value in $(0, 1)$, it samples that value from a $\text{Beta}(\alpha, \beta)$, where $\alpha \equiv \phi\mu$, $\beta \equiv \phi(1 - \mu)$, $\mu \in (0, 1)$, and $\phi \in \mathbb{R}_+$.

The main challenge of using a standard 0-1 inflated beta likelihood is determining how to define $p_{\{0\}}$ and $p_{\{1\}}$. The easiest way to define these parameters is to allow them to be directly estimated. By allowing this, however, we would effectively be injecting multiple discrete components into the model.

A preferable alternative is to use an ordered beta likelihood. Such a likelihood estimates $p_{\{0\}}$ and $p_{\{1\}}$ in terms of the same μ used in the beta component of the model. The idea is that, when μ is close to 1, the mean of the beta component will be close to 1, while $p_{\{1\}}$ will be higher; and when μ is close to 0, the mean of the beta component will be close to 0, and $p_{\{0\}}$ will be higher.

To accomplish this, the ordered beta model defines parameters $p_{\{0\}}$, $p_{(0,1)}$, and $p_{\{1\}}$ via the linked logit model (common from ordinal regression). In addition to μ and ϕ , it is parameterized by *cutpoints* $\mathbf{c} = \langle \mathbf{c}_1, \mathbf{c}_2 \rangle$. $p_{\{0\}}$, $p_{\{1\}}$, and $p_{(0,1)}$ are then defined as follows:

$$\begin{aligned}p_{\{0\}} &\equiv \text{logit}^{-1}(\mathbf{c}_1 - \text{logit}(\mu)) \\ p_{\{1\}} &\equiv \text{logit}^{-1}(\text{logit}(\mu) - \mathbf{c}_2) \\ p_{(0,1)} &\equiv 1 - p_{\{0\}} - p_{\{1\}}\end{aligned}$$

The probability measure is then given by:

$$\mathbb{P}_{\text{OrdBeta}}(x \in (a, b) \mid \mu, \phi, \mathbf{c}) = \begin{cases} 1 & \text{if } a < 0, b > 1 \\ p_{\{0\}} + p_{(0,1)} \int_0^b \text{Beta}(x \mid \mu, \phi) \, dx & \text{if } a < 0, b \leq 1 \\ p_{(0,1)} \int_a^b \text{Beta}(x \mid \mu, \phi) \, dx & \text{if } a \geq 0, b \leq 1 \\ p_{\{1\}} + p_{(0,1)} \int_a^1 \text{Beta}(x \mid \mu, \phi) \, dx & \text{if } a \geq 0, b > 1 \\ 0 & \text{otherwise} \end{cases}$$

The log-likelihood is then:

$$\mathcal{L}_{\text{OrdBeta}}(\mu, \phi, \mathbf{c} \mid \mathbf{x}) = \sum_{i:x_i=0} \log p_{\{0\}} + \sum_{i:x_i=1} \log p_{\{1\}} + \sum_{i:x_i \notin \{0,1\}} \log p_{(0,1)} + \log \text{Beta}(x_i \mid \mu, \phi)$$

Here, $\text{Beta}(x_i \mid \mu, \phi)$ is the beta likelihood. Note that this likelihood can be thought of as a conceptually valid variant of the problematic likelihood described in Appendix A.

We provide models fit using this definition of the likelihood in Appendix C.2.

C Full model specifications

C.1 The truncation models

The models were fit using Stan’s Hamiltonian Markov chain Monte Carlo sampling algorithm. For each model of Degen and Tonhauser’s norming data, we obtained 3,000 posterior samples of the model parameters following 3,000 burn-in samples on four chains, for a total of 12,000 samples. For each model of the factivity data, we obtained 15,000 posterior samples following 15,000 burn-in samples on four chains, for a total of 60,000 samples. For each model of either the bleached or templatic data, we obtained 9,000 posterior samples following 9,000 burn-in samples on four chains, for a total of 36,000 samples. For each model of our replication experiment data, we obtained 45,000 posterior samples with a thinning rate of 3, following 45,000 burn-in samples, on four chains, for a total of 60,000 samples.

C.1.1 Models of the norming data

We encode the certainties for contexts as parameters ω on a log-odds scale, with participant random intercepts ϵ added to these parameters before they are mapped to transformed parameters w for certainty on the unit interval. Normal priors centered at zero are placed on the participant intercepts, as well as the log-odds parameters for contexts; the standard deviations (σ_ϵ and σ_ω) of these normal distributions are, in turn, given exponential hyper-priors. Finally, the likelihood for our model is given by a normal distribution centered at the certainty, whose standard deviation σ_e is parameterized with a prior uniform on the unit interval, truncated to the unit interval. We use $w_{i,j}$ to denote the parameter encoding the certainty for participant j , given context i .

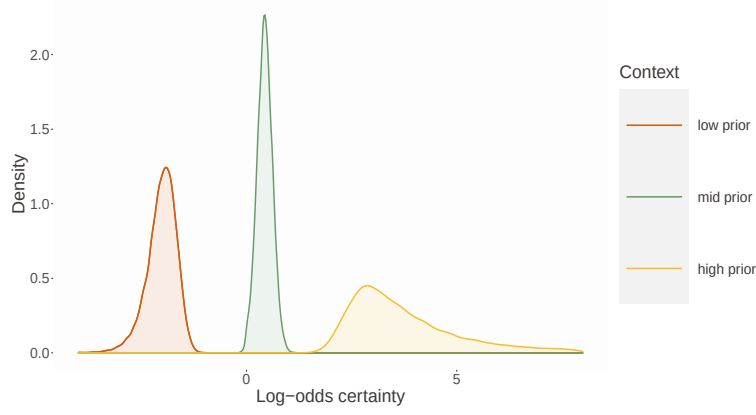


Figure 11: Density plots of the posterior log-odds certainty (with participant intercepts zeroed out) for three items in Degen and Tonhauser’s (2021) norming task. Low and high priors are for *Grace visited her sister*, given the facts *Grace hates her sister* and *Grace loves her sister*, respectively. Mid prior is for *Sophia got a tattoo*, given the fact *Sophia is a hipster*.

The gradient model of the norming data can be defined compactly as follows:²⁷

$$\begin{aligned}
 \sigma_{\omega} &\sim \text{Exponential}(1) \\
 \sigma_{\epsilon_{\omega}} &\sim \text{Exponential}(1) \\
 \sigma_{\epsilon} &\sim \text{Uniform}(0, 1) \\
 \omega &\sim \mathcal{N}(0, \sigma_{\omega}) \\
 \epsilon_{\omega} &\sim \mathcal{N}(0, \sigma_{\epsilon_{\omega}}) \\
 w_{i,j} &= \text{logit}^{-1}(\omega_i + \epsilon_j) \\
 y_{\text{norming}} &\sim \mathcal{N}(w, \sigma_{\epsilon}) \top [0, 1]
 \end{aligned}$$

The parameters ω encode a log-odds certainty rating for each item. We obtain prior distributions for these parameters in our models of factivity by extracting their marginal posterior distributions from the gradient norming model and taking a normal distribution with mean and variance equal to that of this posterior distribution. Density plots for three items are given in Figure 11 (see Figure 14 of Appendix D for all items).

²⁷Throughout all of our implementations, we use non-centered parameterizations of normal distributions. For clarity, we omit this detail in the current presentation.

Meanwhile, the discrete model of the norming data can be presented compactly as follows:

$$\begin{aligned}
\sigma_\omega &\sim \text{Exponential}(1) \\
\sigma_{\epsilon_\omega} &\sim \text{Exponential}(1) \\
\sigma_e &\sim \text{Uniform}(0, 1) \\
\omega &\sim \mathcal{N}(0, \sigma_\omega) \\
\epsilon_\omega &\sim \mathcal{N}(0, \sigma_{\epsilon_\omega}) \\
w_{i,j} &= \text{logit}^{-1}(\omega_i + \epsilon_j) \\
\tau_w &\sim \text{Bernoulli}(w) \\
y_{\text{norming}} &\sim \mathcal{N}(\mathbb{1}(\tau_w), \sigma_e) \text{ T}[0, 1]
\end{aligned}$$

C.1.2 Models of the factivity data

This section provides the four models of factivity and prior world knowledge, which we fit to Degen and Tonhauser’s projection experiment data. In specifying each one, we use μ_ω and σ_ω to denote the means and standard deviations, respectively, of the marginal posterior distributions of the log-odds certainty ratings for the contexts assessed in the norming experiment.

For each model, we encode the log-odds of projection for predicates as parameters ν and the log-odds certainties for contexts as parameters ω . Participant random intercepts ϵ_ν and ϵ_ω are added to these parameters before they are mapped to transformed parameters v and w on the unit interval. Normal priors centered at zero are placed on the participant intercepts, as well as the log-odds parameters for predicates; the standard deviations (σ_{ϵ_ν} , σ_{ϵ_ω} , and σ_ν) of these normals are, in turn, given exponential hyper-priors. Finally, the likelihoods for our models are again given by normal distributions truncated to the unit interval, whose standard deviation σ_e is parameterized by a prior uniform on the unit interval. The mean θ of this truncated normal likelihood varies by model, as we show. In general, we use ‘ $v_{i,k}$ ’ and ‘ $w_{j,k}$ ’ to denote the parameters encoding the probability of projection and certainty, respectively, for participant k , given predicate i and context j .

The discrete-factivity model The discrete-factivity model can be specified as follows:

$$\begin{aligned}
\sigma_v &\sim \text{Exponential}(1) \\
\sigma_{\epsilon_v} &\sim \text{Exponential}(1) \\
\sigma_{\epsilon_\omega} &\sim \text{Exponential}(1) \\
\sigma_e &\sim \text{Uniform}(0, 1) \\
v &\sim \mathcal{N}(0, \sigma_v) \\
\omega &\sim \mathcal{N}(\mu_\omega, \sigma_\omega) \\
\epsilon_v &\sim \mathcal{N}(0, \sigma_{\epsilon_v}) \\
\epsilon_\omega &\sim \mathcal{N}(0, \sigma_{\epsilon_\omega}) \\
\tau_v &\sim \text{Bernoulli}(v) \\
v_{i,k} &= \text{logit}^{-1}(v_i + \epsilon_{vk}) \\
w_{j,k} &= \text{logit}^{-1}(\omega_j + \epsilon_{\omega k}) \\
\theta_{i,j,k} &= \mathbb{1}(\tau_{v_{i,k}}) + \mathbb{1}(\neg \tau_{v_{i,k}}) * w_{j,k} \\
y_{\text{projection}} &\sim \mathcal{N}(\theta, \sigma_e) \top [0, 1]
\end{aligned}$$

Note that adding an anti-veridicality component for each predicate to the discrete-factivity model involves the following adjustments:

$$\begin{aligned}
\sigma_v &\sim \text{Exponential}(1) \\
\sigma_\alpha &\sim \text{Exponential}(1) \\
\sigma_{\epsilon_v} &\sim \text{Exponential}(1) \\
\sigma_{\epsilon_\alpha} &\sim \text{Exponential}(1) \\
\sigma_{\epsilon_\omega} &\sim \text{Exponential}(1) \\
\sigma_e &\sim \text{Uniform}(0, 1) \\
v &\sim \mathcal{N}(0, \sigma_v) \\
\alpha &\sim \mathcal{N}(0, \sigma_\alpha) \\
\omega &\sim \mathcal{N}(\mu_\omega, \sigma_\omega) \\
\epsilon_v &\sim \mathcal{N}(0, \sigma_{\epsilon_v}) \\
\epsilon_\alpha &\sim \mathcal{N}(0, \sigma_{\epsilon_\alpha}) \\
\epsilon_\omega &\sim \mathcal{N}(0, \sigma_{\epsilon_\omega}) \\
\tau_v &\sim \text{Bernoulli}(v) \\
\tau_a &\sim \text{Bernoulli}(a) \\
v_{i,k} &= \text{logit}^{-1}(v_i + \epsilon_{vk}) \\
a_{i,k} &= \text{logit}^{-1}(\alpha_i + \epsilon_{\alpha k}) \\
w_{j,k} &= \text{logit}^{-1}(\omega_j + \epsilon_{\omega k}) \\
\theta_{i,j,k} &= \mathbb{1}(\tau_{v_{i,k}}) + \mathbb{1}(\neg \tau_{v_{i,k}} \wedge \neg \tau_{a_{i,k}}) * w_{j,k} \\
y_{\text{projection}} &\sim \mathcal{N}(\theta, \sigma_e) \top [0, 1]
\end{aligned}$$

The wholly-gradient model The wholly-gradient model can be specified as follows:

$$\begin{aligned}
\sigma_v &\sim \text{Exponential}(1) \\
\sigma_{\epsilon_v} &\sim \text{Exponential}(1) \\
\sigma_{\epsilon_\omega} &\sim \text{Exponential}(1) \\
\sigma_e &\sim \text{Uniform}(0, 1) \\
v &\sim \mathcal{N}(0, \sigma_v) \\
\omega &\sim \mathcal{N}(\mu_\omega, \sigma_\omega) \\
\epsilon_v &\sim \mathcal{N}(0, \sigma_{\epsilon_v}) \\
\epsilon_\omega &\sim \mathcal{N}(0, \sigma_{\epsilon_\omega}) \\
v_{i,k} &= \text{logit}^{-1}(v_i + \epsilon_{vk}) \\
w_{j,k} &= \text{logit}^{-1}(\omega_j + \epsilon_{\omega k}) \\
\theta_{i,j,k} &= v_{i,k} + (1 - v_{i,k}) * w_{j,k} \\
y_{\text{projection}} &\sim \mathcal{N}(\theta, \sigma_e) \text{ T}[0, 1]
\end{aligned}$$

Note that adding an anti-veridicality component for each predicate to the wholly-gradient model involves the following adjustments:

$$\begin{aligned}
\sigma_v &\sim \text{Exponential}(1) \\
\sigma_\alpha &\sim \text{Exponential}(1) \\
\sigma_{\epsilon_v} &\sim \text{Exponential}(1) \\
\sigma_{\epsilon_\alpha} &\sim \text{Exponential}(1) \\
\sigma_{\epsilon_\omega} &\sim \text{Exponential}(1) \\
\sigma_e &\sim \text{Uniform}(0, 1) \\
v &\sim \mathcal{N}(0, \sigma_v) \\
\alpha &\sim \mathcal{N}(0, \sigma_\alpha) \\
\omega &\sim \mathcal{N}(\mu_\omega, \sigma_\omega) \\
\epsilon_v &\sim \mathcal{N}(0, \sigma_{\epsilon_v}) \\
\epsilon_\alpha &\sim \mathcal{N}(0, \sigma_{\epsilon_\alpha}) \\
\epsilon_\omega &\sim \mathcal{N}(0, \sigma_{\epsilon_\omega}) \\
v_{i,k} &= \text{logit}^{-1}(v_i + \epsilon_{vk}) \\
a_{i,k} &= \text{logit}^{-1}(\alpha_i + \epsilon_{\alpha k}) \\
w_{j,k} &= \text{logit}^{-1}(\omega_j + \epsilon_{\omega k}) \\
\theta_{i,j,k} &= v_{i,k} + (1 - v_{i,k}) * (1 - a_{i,k}) * w_{j,k} \\
y_{\text{projection}} &\sim \mathcal{N}(\theta, \sigma_e) \text{ T}[0, 1]
\end{aligned}$$

The discrete-world model The discrete-world model is defined similarly to the discrete-factivity model, except by alternating which parameters are taken to make discrete versus gradient contributions to the response. Now, world knowledge affects the certainty discretely, producing values

of either 0 or 1. Meanwhile, if the certainty is 0, the factivity of the relevant predicate makes a gradient contribution to the response.

$$\begin{aligned}
\sigma_v &\sim \text{Exponential}(1) \\
\sigma_{\epsilon_v} &\sim \text{Exponential}(1) \\
\sigma_{\epsilon_\omega} &\sim \text{Exponential}(1) \\
\sigma_e &\sim \text{Uniform}(0, 1) \\
v &\sim \mathcal{N}(0, \sigma_v) \\
\omega &\sim \mathcal{N}(\mu_\omega, \sigma_\omega) \\
\epsilon_v &\sim \mathcal{N}(0, \sigma_{\epsilon_v}) \\
\epsilon_\omega &\sim \mathcal{N}(0, \sigma_{\epsilon_\omega}) \\
\tau_w &\sim \text{Bernoulli}(w) \\
v_{i,k} &= \text{logit}^{-1}(v_i + \epsilon_{vk}) \\
w_{j,k} &= \text{logit}^{-1}(\omega_j + \epsilon_{\omega k}) \\
\theta_{i,j,k} &= \mathbb{1}(\tau_{w_{j,k}}) + \mathbb{1}(\neg \tau_{w_{j,k}}) * v_{i,k} \\
y_{\text{projection}} &\sim \mathcal{N}(\theta, \sigma_e) \top [0, 1]
\end{aligned}$$

Note that adding an anti-veridicality component for each predicate to the discrete-world model does not change it.

The wholly-discrete model Finally, the wholly-discrete model generates parameters θ which are either 0 or 1, depending on two Bernoullis parameterized by the probabilities of projection and world-knowledge-derived certainties, respectively. Each parameter $\theta_{i,j,k}$ is thus 0 with probability $p = (1 - v_{i,k}) * (1 - w_{j,k})$, and 1 with probability $1 - p$.

$$\begin{aligned}
\sigma_v &\sim \text{Exponential}(1) \\
\sigma_{\epsilon_v} &\sim \text{Exponential}(1) \\
\sigma_{\epsilon_\omega} &\sim \text{Exponential}(1) \\
\sigma_e &\sim \text{Uniform}(0, 1) \\
v &\sim \mathcal{N}(0, \sigma_v) \\
\omega &\sim \mathcal{N}(\mu_\omega, \sigma_\omega) \\
\epsilon_v &\sim \mathcal{N}(0, \sigma_{\epsilon_v}) \\
\epsilon_\omega &\sim \mathcal{N}(0, \sigma_{\epsilon_\omega}) \\
\tau_v &\sim \text{Bernoulli}(v) \\
\tau_w &\sim \text{Bernoulli}(w) \\
v_{i,k} &= \text{logit}^{-1}(v_i + \epsilon_{vk}) \\
w_{j,k} &= \text{logit}^{-1}(\omega_j + \epsilon_{\omega k}) \\
\theta_{i,j,k} &= \mathbb{1}(\tau_{v_{i,k}} \vee \tau_{w_{j,k}}) \\
y_{\text{projection}} &\sim \mathcal{N}(\theta, \sigma_e) \top [0, 1]
\end{aligned}$$

Note that adding an anti-veridicality component for each predicate to the wholly-discrete model involves the following adjustments:

$$\begin{aligned}
\sigma_v &\sim \text{Exponential}(1) \\
\sigma_\alpha &\sim \text{Exponential}(1) \\
\sigma_{\epsilon_v} &\sim \text{Exponential}(1) \\
\sigma_{\epsilon_\alpha} &\sim \text{Exponential}(1) \\
\sigma_{\epsilon_\omega} &\sim \text{Exponential}(1) \\
\sigma_e &\sim \text{Uniform}(0, 1) \\
v &\sim \mathcal{N}(0, \sigma_v) \\
\alpha &\sim \mathcal{N}(0, \sigma_\alpha) \\
\omega &\sim \mathcal{N}(\mu_\omega, \sigma_\omega) \\
\epsilon_v &\sim \mathcal{N}(0, \sigma_{\epsilon_v}) \\
\epsilon_\alpha &\sim \mathcal{N}(0, \sigma_{\epsilon_\alpha}) \\
\epsilon_\omega &\sim \mathcal{N}(0, \sigma_{\epsilon_\omega}) \\
\tau_v &\sim \text{Bernoulli}(v) \\
\tau_a &\sim \text{Bernoulli}(a) \\
\tau_w &\sim \text{Bernoulli}(w) \\
v_{i,k} &= \text{logit}^{-1}(v_i + \epsilon_{v_k}) \\
a_{i,k} &= \text{logit}^{-1}(\alpha_i + \epsilon_{\alpha_k}) \\
w_{j,k} &= \text{logit}^{-1}(\omega_j + \epsilon_{\omega_k}) \\
\theta_{i,j,k} &= \mathbb{1}(\tau_{v_{i,k}} \vee (\neg \tau_{a_{i,k}} \wedge \tau_{w_{j,k}})) \\
y_{\text{projection}} &\sim \mathcal{N}(\theta, \sigma_e) \text{ T}[0, 1]
\end{aligned}$$

C.1.3 Models of the contentful evaluation data

To evaluate the four models using this data, we obtained, from each model, means μ_v and standard deviations σ_v of the marginal posterior distributions over the log-odds of projection for predicates, as well as means μ_ω and standard deviations σ_ω of the marginal posterior distributions over the log-odds certainty for contexts. We then used normal distributions with these means and standard deviations as prior distributions for the corresponding parameters in the models constructed for the evaluations. (See Figure 12 for density plots of these posterior distributions for six predicates, and Figure 15 of Appendix D for density plots of the posterior distributions for all predicates.)

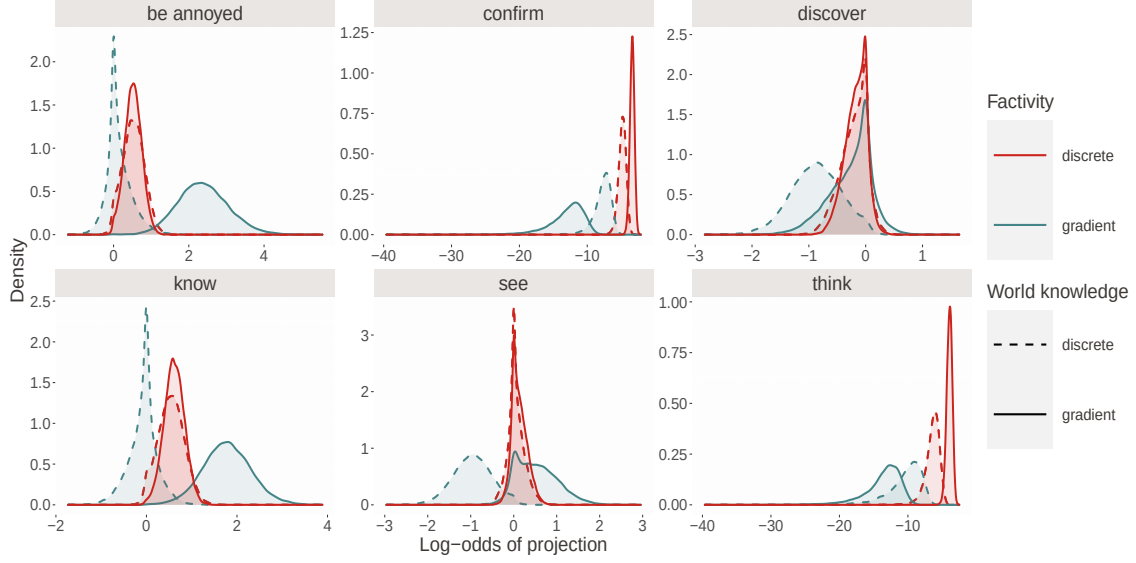


Figure 12: Density plots of the posterior log-odds of projection (with participant intercepts zeroed out) for all four models for six predicates from Degen and Tonhauser’s (2021) projection experiment.

Each model of this evaluation data has the following structure:

$$\begin{aligned}
 \sigma_{\epsilon_v} &\sim \text{Exponential}(1) \\
 \sigma_{\epsilon_\omega} &\sim \text{Exponential}(1) \\
 \sigma_e &\sim \text{Uniform}(0, 1) \\
 \nu &\sim \mathcal{N}(\mu_\nu, \sigma_\nu) \\
 \omega &\sim \mathcal{N}(\mu_\omega, \sigma_\omega) \\
 \epsilon_v &\sim \mathcal{N}(0, \sigma_{\epsilon_v}) \\
 \epsilon_\omega &\sim \mathcal{N}(0, \sigma_{\epsilon_\omega}) \\
 &\vdots \\
 v_{i,k} &= \text{logit}^{-1}(\nu_i + \epsilon_{vk}) \\
 w_{j,k} &= \text{logit}^{-1}(\omega_j + \epsilon_{\omega k}) \\
 \theta_{i,j,k} &= \dots \\
 y_{\text{replication}} &\sim \mathcal{N}(\theta, \sigma_e) \text{ T}[0, 1]
 \end{aligned}$$

The ellipses are used to represent the parts specific to each model. For example, the specification of the discrete-factivity model would include the sampling statement ‘ $\tau_v \sim \text{Bernoulli}(v)$ ’, and the definition of $\theta_{i,j,k}$ would be $\mathbb{1}(\tau_{v_{i,j,k}}) + \mathbb{1}(\neg \tau_{v_{i,j,k}}) * w_{j,k}$.

C.1.4 Models of the non-contentful evaluation data

To evaluate the four models in terms of the bleached and templatic data, we used the means μ_ν and standard deviations σ_ν of the marginal posterior distributions over the log-odds of projection

that we used for the evaluations on the replication experiment data. As before, we use normal distributions with these means and standard deviations as prior distributions for the corresponding parameters in the models constructed for the evaluations. Then, in each evaluation, we inferred a distribution over the parameters σ_ω and ω that encode the certainty associated with either the bleached or the templatic context.

In particular, both the bleached and the templatic evaluations have the following structure, where ellipses, as above, indicate the unique aspects of each of the four models:

$$\begin{aligned}
\sigma_\omega &\sim \text{Exponential}(1) \\
\sigma_{\epsilon_v} &\sim \text{Exponential}(1) \\
\sigma_{\epsilon_\omega} &\sim \text{Exponential}(1) \\
\sigma_e &\sim \text{Uniform}(0, 1) \\
\mathbf{v} &\sim \mathcal{N}(\boldsymbol{\mu}_v, \boldsymbol{\sigma}_v) \\
\omega &\sim \mathcal{N}(0, \sigma_\omega) \\
\epsilon_v &\sim \mathcal{N}(0, \sigma_{\epsilon_v}) \\
\epsilon_\omega &\sim \mathcal{N}(0, \sigma_{\epsilon_\omega}) \\
&\vdots \\
\mathbf{v}_{i,j} &= \text{logit}^{-1}(\mathbf{v}_i + \epsilon_{v,j}) \\
\mathbf{w}_j &= \text{logit}^{-1}(\omega + \epsilon_{\omega,j}) \\
\boldsymbol{\theta}_{i,j} &= \dots \\
\mathbf{y}_{\text{non-contentful}} &\sim \mathcal{N}(\boldsymbol{\theta}, \sigma_e) \top [0, 1]
\end{aligned}$$

The data tuple $\mathbf{y}_{\text{non-contentful}}$ should be understood as either $\mathbf{y}_{\text{bleached}}$ or $\mathbf{y}_{\text{templatic}}$, depending on the evaluation performed.

C.2 The inflation models

The same general procedure was followed to fit the inflation models: first, we fit models of Degen and Tonhauser’s norming data; then we extracted posterior distributions over the model parameters, which we used to inform our priors for the models of their factivity data. We again fit both gradient and discrete models of the norming data, obtaining 3,000 posterior samples of the model parameters following 3,000 burn-in samples on four chains, for a total of 12,000 samples. For the factivity models, we obtained 6,000 posterior samples following 3,000 burn-in samples on four chains, for a total of 24,000 samples.

In general, the inflation models are identical to their truncation counterparts; the only difference is the definition of the likelihood.

C.2.1 Models of the norming data

The gradient model of the norming data can be presented compactly as follows:

$$\begin{aligned}
\sigma_{\omega} &\sim \text{Exponential}(1) \\
\sigma_{\epsilon_{\omega}} &\sim \text{Exponential}(1) \\
\sigma_{\epsilon_{\kappa,1}} &\sim \text{Exponential}(1) \\
\sigma_{\epsilon_{\kappa,2}} &\sim \text{Exponential}(1) \\
\kappa &\sim \mathcal{N}(\log(4), 1) \\
\phi &\sim \text{Exponential}(0.1) \\
\omega &\sim \mathcal{N}(0, \sigma_{\omega}) \\
\epsilon_{\omega} &\sim \mathcal{N}(0, \sigma_{\epsilon_{\omega}}) \\
\epsilon_{\kappa,1} &\sim \mathcal{N}(0, \sigma_{\epsilon_{\kappa,1}}) \\
\epsilon_{\kappa,2} &\sim \mathcal{N}(0, \sigma_{\epsilon_{\kappa,2}}) \\
w_{i,j} &= \text{logit}^{-1}(\omega_i + \epsilon_{\omega j}) \\
c_{1,i,j} &= -e^{\kappa + \epsilon_{\kappa,1} j} \\
c_{2,i,j} &= e^{\kappa + \epsilon_{\kappa,2} j} \\
y_{\text{norming}} &\sim \text{OrdBeta}(w, \phi, c)
\end{aligned}$$

The parameters ω encode a log-odds certainty rating for each item. Following our strategy for the truncation models, we obtain prior distributions for these parameters in our models of factivity by extracting their marginal posterior distributions from our norming model and, for each item (i.e., each parameter of ω), taking a normal distribution with mean and variance equal to that of its posterior distribution in the norming model.

Meanwhile, the discrete model of the norming data is the following:

$$\begin{aligned}
\sigma_{\omega} &\sim \text{Exponential}(1) \\
\sigma_{\epsilon_{\omega}} &\sim \text{Exponential}(1) \\
\sigma_{\epsilon_{\kappa,1}} &\sim \text{Exponential}(1) \\
\sigma_{\epsilon_{\kappa,2}} &\sim \text{Exponential}(1) \\
\kappa &\sim \mathcal{N}(\log(4), 1) \\
\phi &\sim \text{Exponential}(0.1) \\
\eta &\sim \mathcal{N}(1.5, 1) \top [0,] \\
\omega &\sim \mathcal{N}(0, \sigma_{\omega}) \\
\epsilon_{\omega} &\sim \mathcal{N}(0, \sigma_{\epsilon_{\omega}}) \\
\epsilon_{\kappa,1} &\sim \mathcal{N}(0, \sigma_{\epsilon_{\kappa,1}}) \\
\epsilon_{\kappa,2} &\sim \mathcal{N}(0, \sigma_{\epsilon_{\kappa,2}}) \\
\tau_w &\sim \text{Bernoulli}(w) \\
w_{i,j} &= \text{logit}^{-1}(\omega_i + \epsilon_{\omega j}) \\
c_{1,i,j} &= -e^{\kappa + \epsilon_{\kappa,1}j} \\
c_{2,i,j} &= e^{\kappa + \epsilon_{\kappa,2}j} \\
\mu_1 &= \text{logit}^{-1}(\eta) \\
\mu_0 &= 1 - \text{logit}^{-1}(\eta) \\
\theta_{i,j} &= \mathbb{1}(\tau_w)_{i,j} * \mu_1 + \mathbb{1}(\neg \tau_w)_{i,j} * \mu_0 \\
\mathbf{y}_{\text{norming}} &\sim \text{OrdBeta}(\theta, \phi, \mathbf{c})
\end{aligned}$$

C.2.2 Models of the factivity data

The discrete-factivity model

$$\begin{aligned}
\sigma_v &\sim \text{Exponential}(1) \\
\sigma_{\epsilon_v} &\sim \text{Exponential}(1) \\
\sigma_{\epsilon_\omega} &\sim \text{Exponential}(1) \\
\sigma_{\epsilon_{\kappa,1}} &\sim \text{Exponential}(1) \\
\sigma_{\epsilon_{\kappa,2}} &\sim \text{Exponential}(1) \\
\kappa &\sim \mathcal{N}(\log(4), 1) \\
\phi &\sim \text{Exponential}(0.1) \\
\eta &\sim \mathcal{N}(1.5, 1) \text{ T}[0,] \\
v &\sim \mathcal{N}(0, \sigma_v) \\
\omega &\sim \mathcal{N}(\mu_\omega, \sigma_\omega) \\
\epsilon_v &\sim \mathcal{N}(0, \sigma_{\epsilon_v}) \\
\epsilon_\omega &\sim \mathcal{N}(0, \sigma_{\epsilon_\omega}) \\
\epsilon_{\kappa,1} &\sim \mathcal{N}(0, \sigma_{\epsilon_{\kappa,1}}) \\
\epsilon_{\kappa,2} &\sim \mathcal{N}(0, \sigma_{\epsilon_{\kappa,2}}) \\
\tau_v &\sim \text{Bernoulli}(v) \\
v_{i,k} &= \text{logit}^{-1}(v_i + \epsilon_{vk}) \\
w_{j,k} &= \text{logit}^{-1}(\omega_j + \epsilon_{\omega k}) \\
c_{1,i,j,k} &= -e^{\kappa + \epsilon_{\kappa,1k}} \\
c_{2,i,j,k} &= e^{\kappa + \epsilon_{\kappa,2k}} \\
\mu_1 &= \text{logit}^{-1}(\eta) \\
\theta_{i,j,k} &= \mathbb{1}(\tau_{v_{i,k}}) * \mu_1 + \mathbb{1}(\neg \tau_{v_{i,k}}) * w_{j,k} \\
y_{\text{projection}} &\sim \text{OrdBeta}(\theta, \phi, c)
\end{aligned}$$

The wholly-gradient model

$$\begin{aligned}
\sigma_v &\sim \text{Exponential}(1) \\
\sigma_{\epsilon_v} &\sim \text{Exponential}(1) \\
\sigma_{\epsilon_\omega} &\sim \text{Exponential}(1) \\
\sigma_{\epsilon_{\kappa,1}} &\sim \text{Exponential}(1) \\
\sigma_{\epsilon_{\kappa,2}} &\sim \text{Exponential}(1) \\
\kappa &\sim \mathcal{N}(\log(4), 1) \\
\phi &\sim \text{Exponential}(0.1) \\
v &\sim \mathcal{N}(0, \sigma_v) \\
\omega &\sim \mathcal{N}(\mu_\omega, \sigma_\omega) \\
\epsilon_v &\sim \mathcal{N}(0, \sigma_{\epsilon_v}) \\
\epsilon_\omega &\sim \mathcal{N}(0, \sigma_{\epsilon_\omega}) \\
\epsilon_{\kappa,1} &\sim \mathcal{N}(0, \sigma_{\epsilon_{\kappa,1}}) \\
\epsilon_{\kappa,2} &\sim \mathcal{N}(0, \sigma_{\epsilon_{\kappa,2}}) \\
v_{i,k} &= \text{logit}^{-1}(v_i + \epsilon_{vk}) \\
w_{j,k} &= \text{logit}^{-1}(\omega_j + \epsilon_{\omega k}) \\
c_{1,i,j,k} &= -e^{\kappa + \epsilon_{\kappa,1k}} \\
c_{2,i,j,k} &= e^{\kappa + \epsilon_{\kappa,2k}} \\
\theta_{i,j,k} &= v_{i,k} + (1 - v_{i,k}) * w_{j,k} \\
y_{\text{projection}} &\sim \text{OrdBeta}(\theta, \phi, c)
\end{aligned}$$

The discrete-world model

$$\begin{aligned}
\sigma_{\nu} &\sim \text{Exponential}(1) \\
\sigma_{\epsilon_{\nu}} &\sim \text{Exponential}(1) \\
\sigma_{\epsilon_{\omega}} &\sim \text{Exponential}(1) \\
\sigma_{\epsilon_{\kappa,1}} &\sim \text{Exponential}(1) \\
\sigma_{\epsilon_{\kappa,2}} &\sim \text{Exponential}(1) \\
\kappa &\sim \mathcal{N}(\log(4), 1) \\
\phi &\sim \text{Exponential}(0.1) \\
\eta &\sim \mathcal{N}(1.5, 1) \text{ T}[0,] \\
\nu &\sim \mathcal{N}(0, \sigma_{\nu}) \\
\omega &\sim \mathcal{N}(\mu_{\omega}, \sigma_{\omega}) \\
\epsilon_{\nu} &\sim \mathcal{N}(0, \sigma_{\epsilon_{\nu}}) \\
\epsilon_{\omega} &\sim \mathcal{N}(0, \sigma_{\epsilon_{\omega}}) \\
\epsilon_{\kappa,1} &\sim \mathcal{N}(0, \sigma_{\epsilon_{\kappa,1}}) \\
\epsilon_{\kappa,2} &\sim \mathcal{N}(0, \sigma_{\epsilon_{\kappa,2}}) \\
\tau_w &\sim \text{Bernoulli}(w) \\
v_{i,k} &= \text{logit}^{-1}(\nu_i + \epsilon_{\nu k}) \\
w_{j,k} &= \text{logit}^{-1}(\omega_j + \epsilon_{\omega k}) \\
c_{1,i,j,k} &= -e^{\kappa + \epsilon_{\kappa,1k}} \\
c_{2,i,j,k} &= e^{\kappa + \epsilon_{\kappa,2k}} \\
\mu_1 &= \text{logit}^{-1}(\eta) \\
\theta_{i,j,k} &= \mathbb{1}(\tau_{w_{j,k}}) * \mu_1 + \mathbb{1}(\neg \tau_{w_{j,k}}) * v_{i,k} \\
y_{\text{projection}} &\sim \text{OrdBeta}(\theta, \phi, c)
\end{aligned}$$

The wholly-discrete model

$$\begin{aligned}
\sigma_v &\sim \text{Exponential}(1) \\
\sigma_{\epsilon_v} &\sim \text{Exponential}(1) \\
\sigma_{\epsilon_\omega} &\sim \text{Exponential}(1) \\
\sigma_{\epsilon_{\kappa,1}} &\sim \text{Exponential}(1) \\
\sigma_{\epsilon_{\kappa,2}} &\sim \text{Exponential}(1) \\
\kappa &\sim \mathcal{N}(\log(4), 1) \\
\phi &\sim \text{Exponential}(0.1) \\
\eta &\sim \mathcal{N}(1.5, 1) \top [0,] \\
\nu &\sim \mathcal{N}(0, \sigma_v) \\
\omega &\sim \mathcal{N}(\mu_\omega, \sigma_\omega) \\
\epsilon_v &\sim \mathcal{N}(0, \sigma_{\epsilon_v}) \\
\epsilon_\omega &\sim \mathcal{N}(0, \sigma_{\epsilon_\omega}) \\
\epsilon_{\kappa,1} &\sim \mathcal{N}(0, \sigma_{\epsilon_{\kappa,1}}) \\
\epsilon_{\kappa,2} &\sim \mathcal{N}(0, \sigma_{\epsilon_{\kappa,2}}) \\
\tau_w &\sim \text{Bernoulli}(w) \\
v_{i,k} &= \text{logit}^{-1}(\nu_i + \epsilon_{v_k}) \\
w_{j,k} &= \text{logit}^{-1}(\omega_j + \epsilon_{w_k}) \\
c_{1,i,j,k} &= -e^{\kappa + \epsilon_{\kappa,1k}} \\
c_{2,i,j,k} &= e^{\kappa + \epsilon_{\kappa,2k}} \\
\mu_1 &= \text{logit}^{-1}(\eta) \\
\mu_0 &= 1 - \text{logit}^{-1}(\eta) \\
\theta_{i,j,k} &= \mathbb{1}(\tau_{v_{i,k}} \vee \tau_{w_{j,k}}) * \mu_1 + \mathbb{1}(\neg(\tau_{v_{i,k}} \vee \tau_{w_{j,k}})) * \mu_0 \\
y_{\text{projection}} &\sim \text{OrdBeta}(\theta, \phi, c)
\end{aligned}$$

C.2.3 Comparisons

Figure 13 (left) presents ELPDs for the four models. Again, the discrete model performs substantially better than any of the other models; indeed, we obtain the same ranking as we did for the truncation models. Figure 13 (right) presents ELPDs for all four models—both the truncation and the inflation variants—on the same dataset, but excluding the endpoints (which are assigned mass, rather than density, likelihoods by the inflation models). Remarkably, each truncation model performs substantially better than its inflation counterpart on this subset of the data; moreover, the worst-performing truncation model (discrete-world) still performs numerically better than the best-performing inflation model (discrete-factivity).

We can therefore conclude two things. First, our initial comparison in terms of truncated normal distributions is unlikely to favor the discrete-factivity model due to its particular choice of likelihood. Second, the truncated normal likelihood, in fact, provides better models of the data than the ordered beta likelihood presented here.

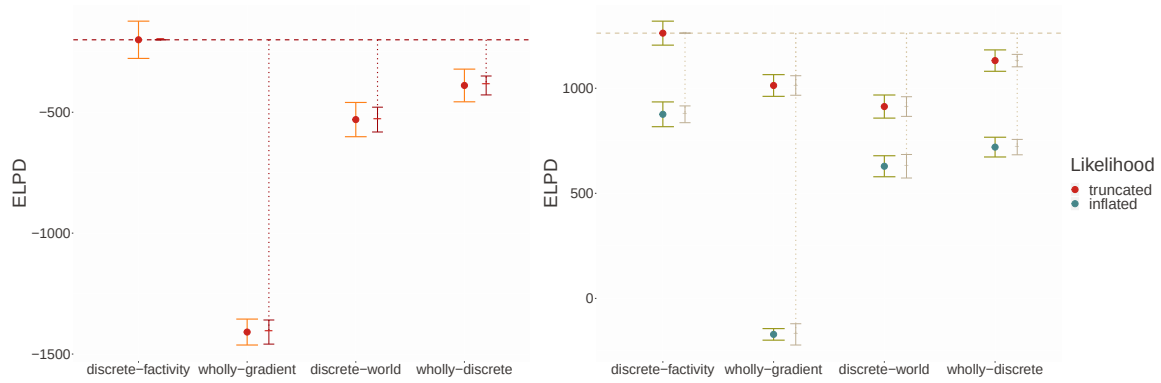


Figure 13: Left: ELPDs for all four ordered beta models. Right: ELPDs for all four truncated normal models, along with their ordered beta counterparts, on the subset of Degen and Tonhauser’s data that excludes 0 and 1 responses. Dotted lines indicate estimated differences between each model and the discrete-factivity model. Error bars indicate standard errors.

D Plots

D.1 Posterior parameter distributions

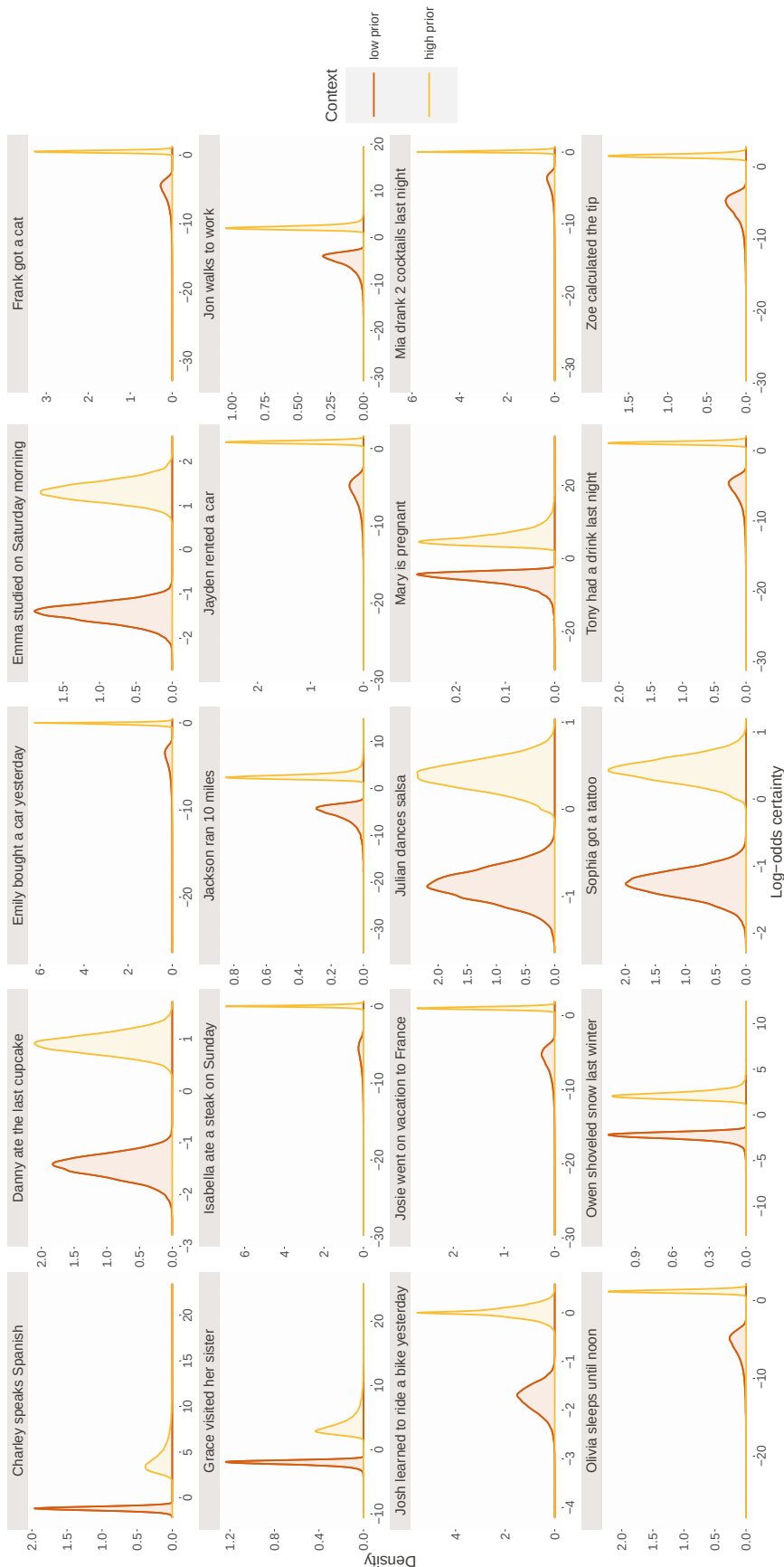


Figure 14: Density plots of the posterior log-odds certainty (with participant intercepts zeroed out) for all items in Degen and Tonhauser's (2021) norming task.

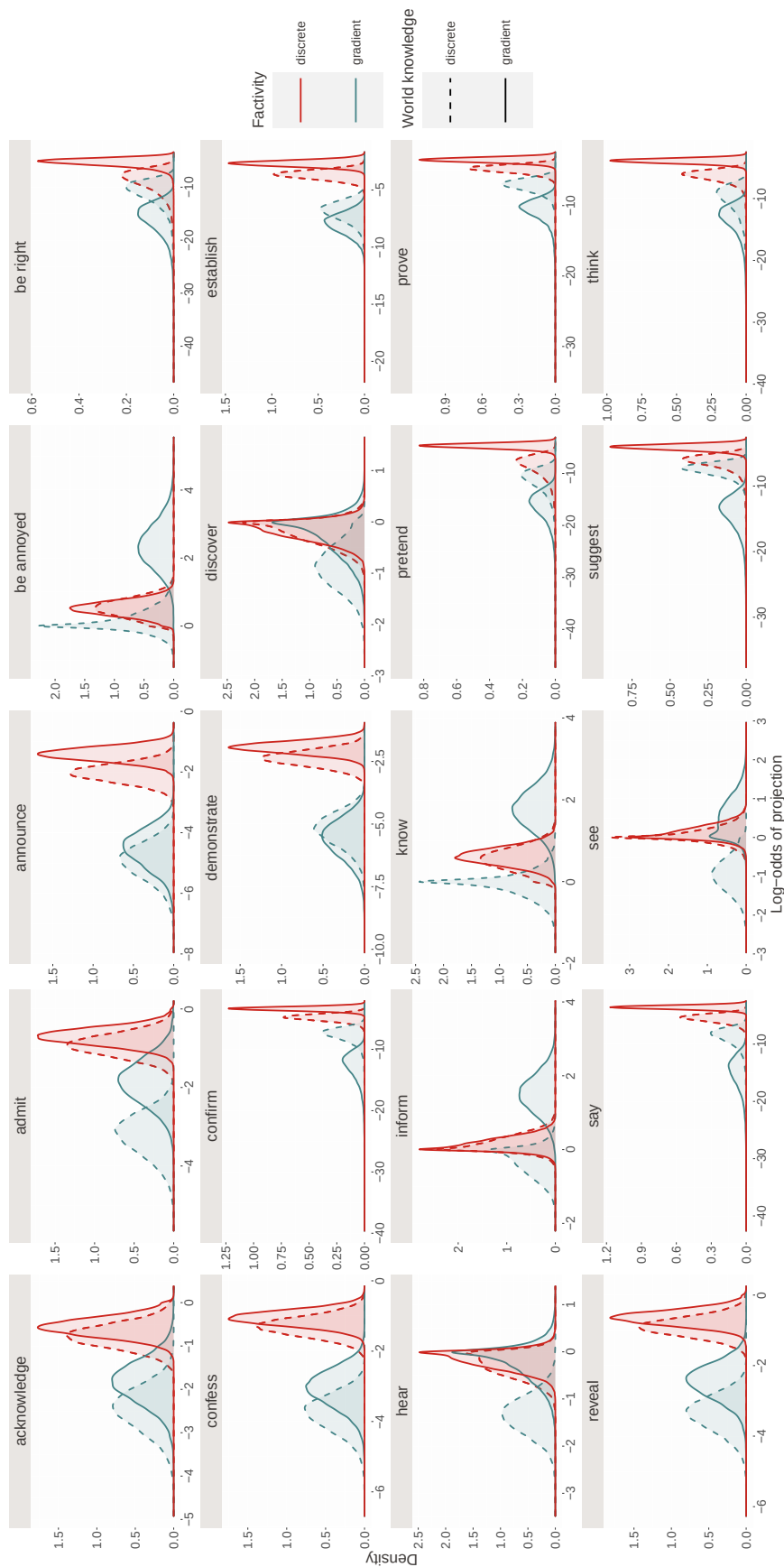


Figure 15: Density plots of the posterior log-odds of projection (with participant intercepts zeroed out) for all four models for all predicates in Degen and Tonhauser’s (2021) projection experiment.

D.2 Posterior predictive distributions

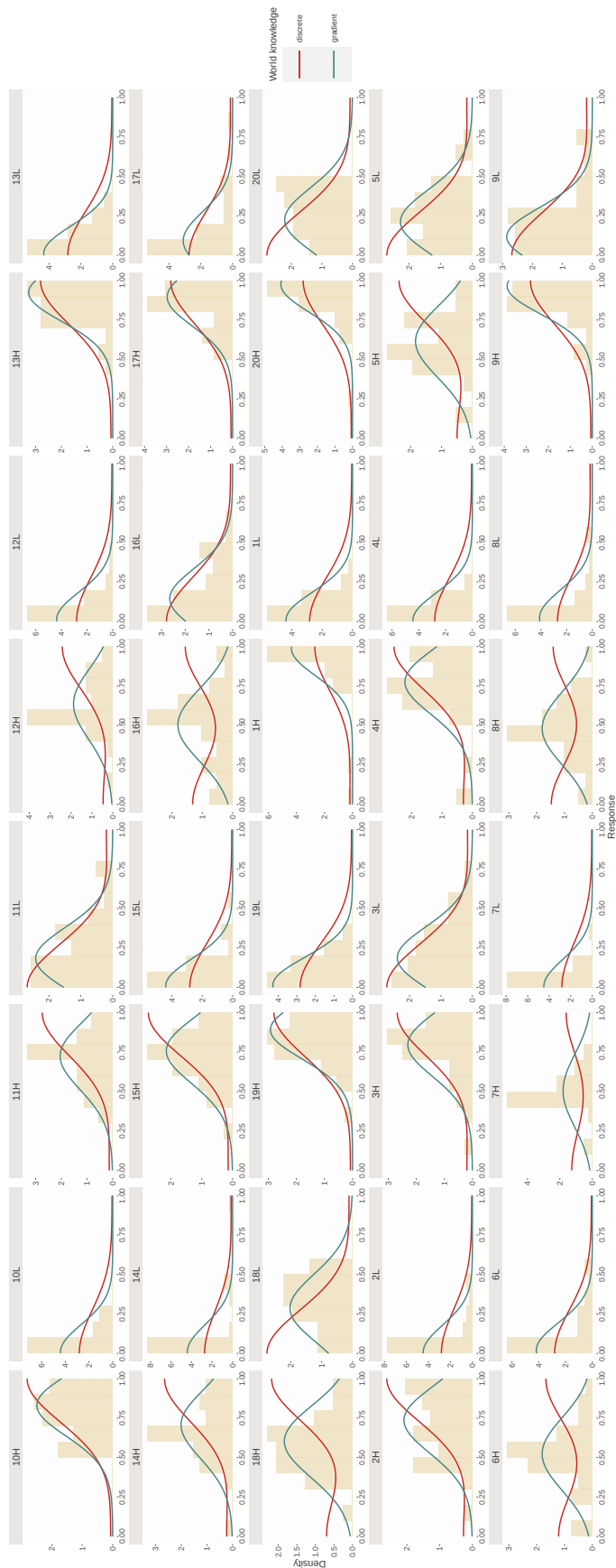


Figure 16: Posterior predictive distributions (with simulated participant intercepts) of the norming-gradient and norming-discrete models for all items in Degen and Tonhauser’s (2021) norming experiment. Empirical distributions are represented by density histograms of data from Degen and Tonhauser 2021.

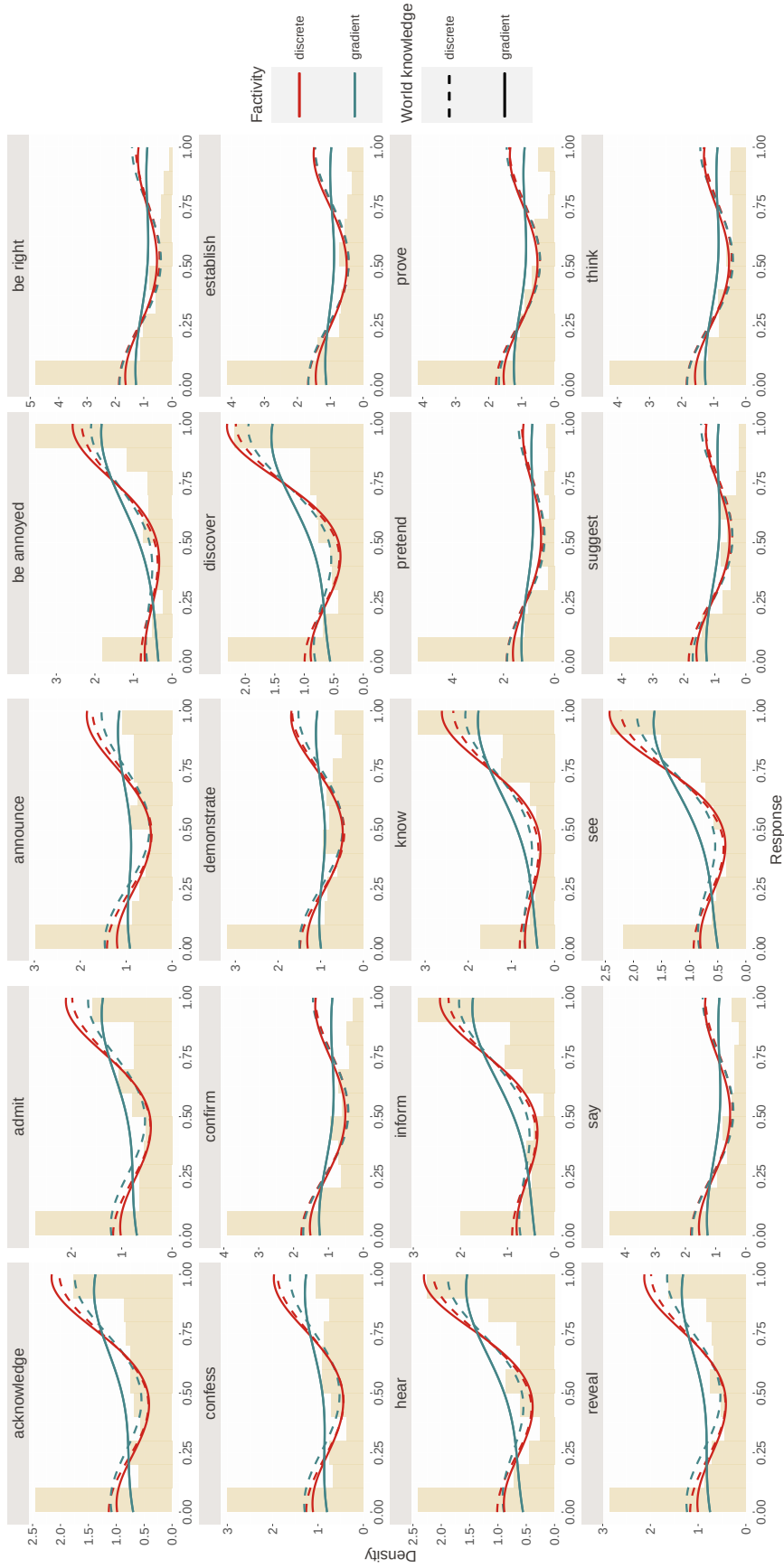


Figure 17: Posterior predictive distributions (with simulated participant intercepts) of all four models for all predicates in Degen and Tonhauser’s (2021) projection experiment, for all contexts combined. Empirical distributions are represented by density histograms of data from Degen and Tonhauser 2021.

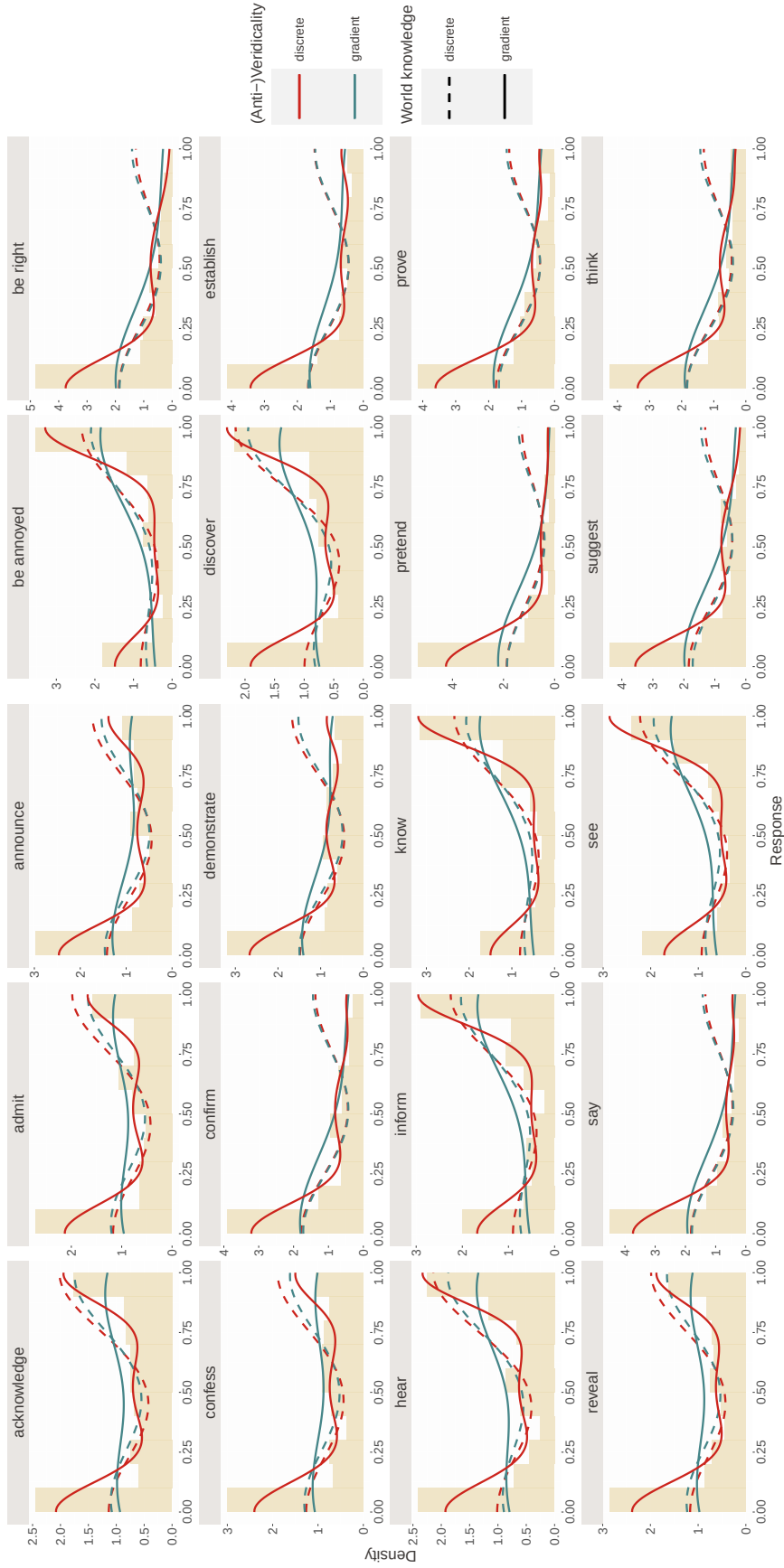


Figure 18: Posterior predictive distributions (with simulated participant intercepts) of all four models for all predicates in Degen and Tonhauser’s (2021) projection experiment, for all contexts combined. Here each predicate may also take on an anti-veridical interpretation. Empirical distributions are represented by density histograms of data from Degen and Tonhauser 2021.

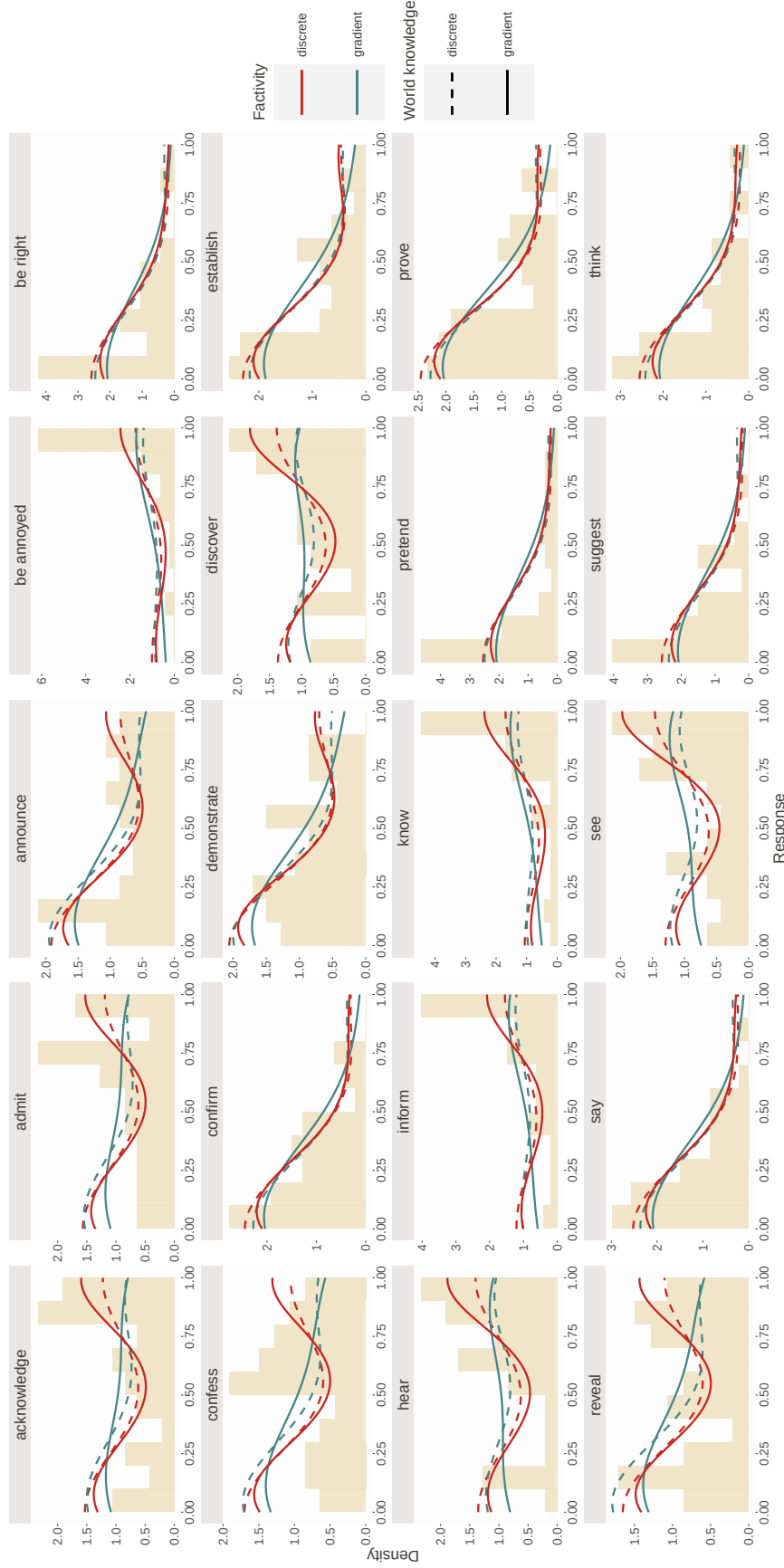


Figure 19: Posterior predictive distributions (with simulated participant intercepts) of all four model evaluations for all predicates in (3). Complement clause: *a particular thing happened*. Empirical distributions are represented by density histograms.

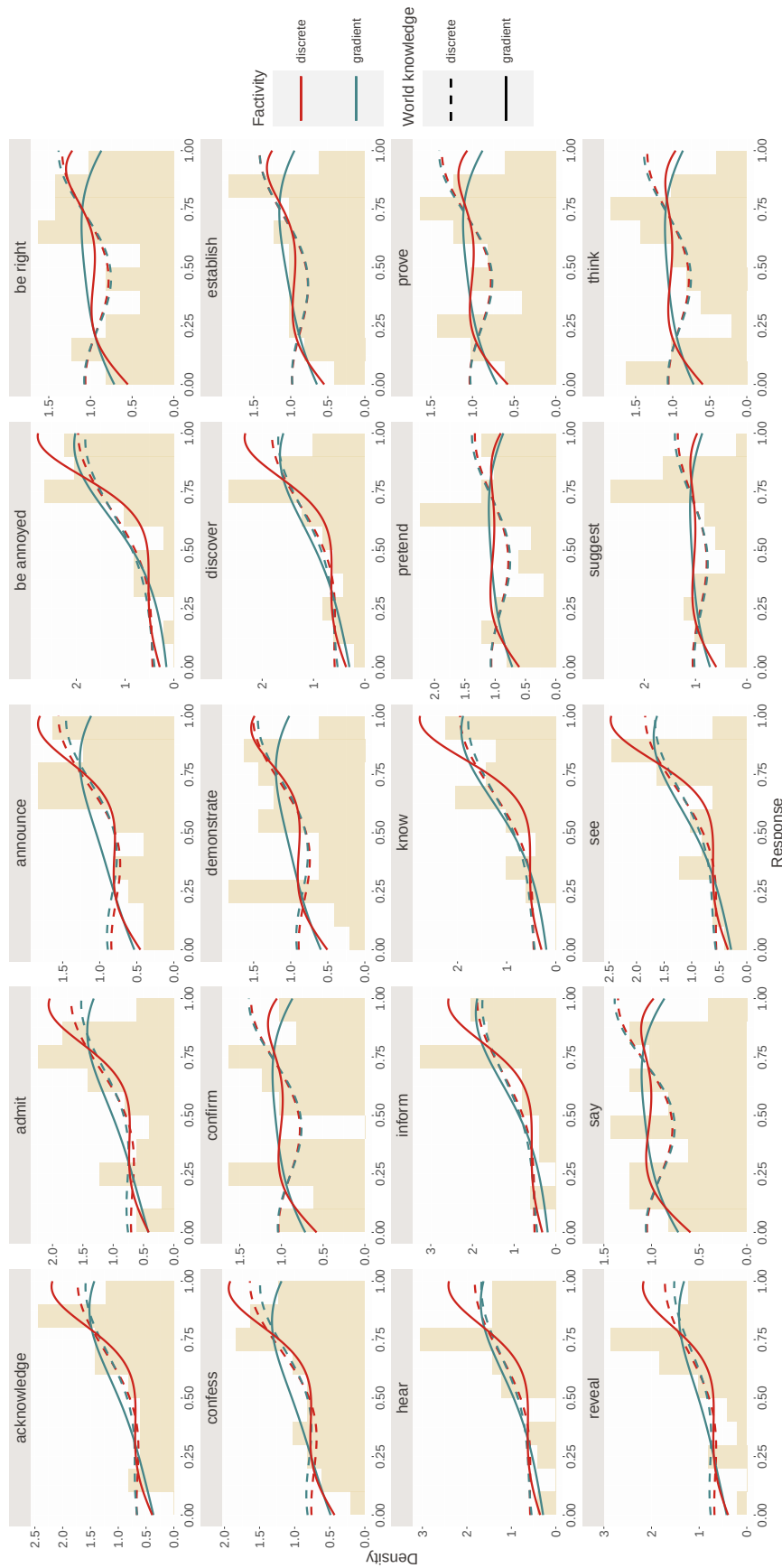


Figure 20: Posterior predictive distributions (with simulated participant intercepts) of all four model evaluations for all predicates in (3). Complement clause: *X happened*. Empirical distributions are represented by density histograms.